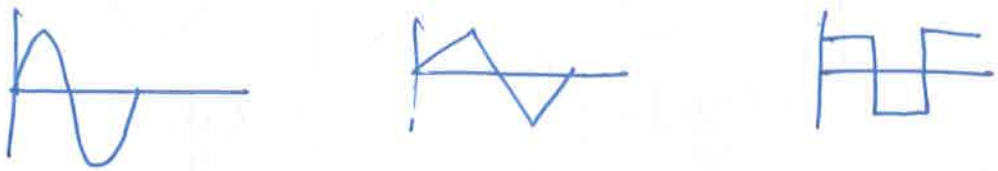
ALTERNATING WAVEFORMS

Alternating over time b/w two prescribed levels.

- Sinusoidal, triangular, pulse

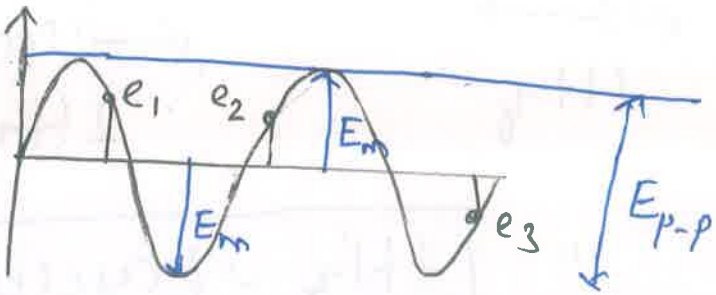


Point of interest in this course is Sinusoidal ac Voltage and current.

Generation:- Generation of ac Voltage may be by ① ac generator ② functional generators, ③ inverters.

Definitions:-

(a) waveform:-



$A = f(t)$

↳ time, position, degree, radian, temp....

↳ Voltage

(b) Instantaneous value:- e_1, e_2, e_3.

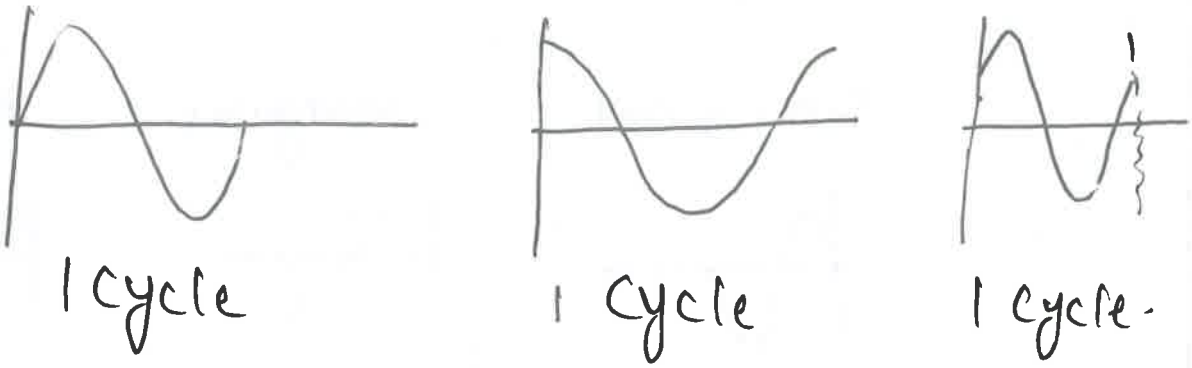
$v = V_m \sin\theta$ or $e = E_m \sin\omega t$

(c) Peak amplitude:- E_m.

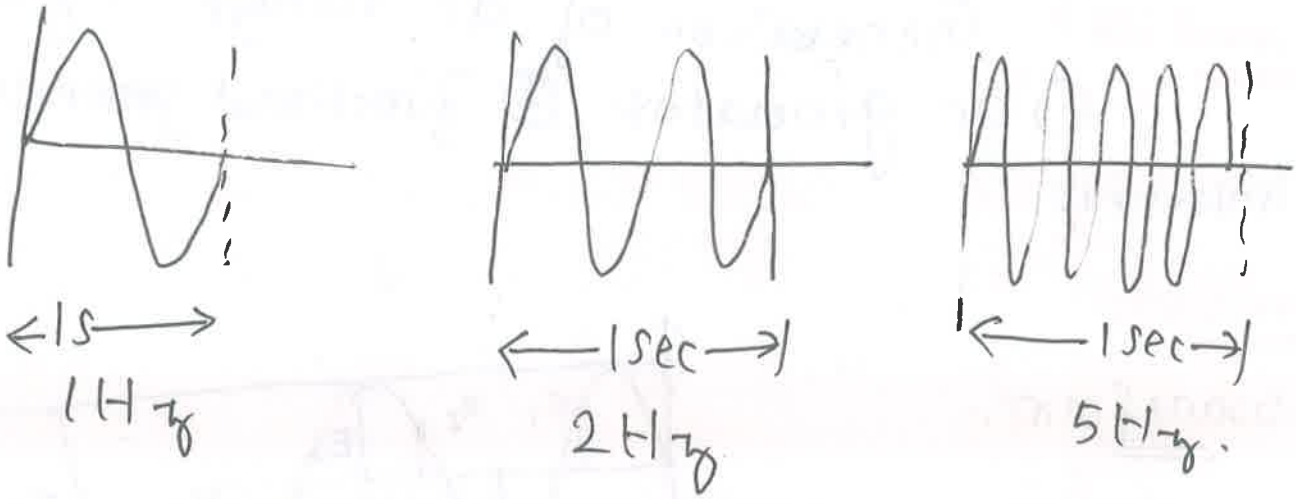
(d) Peak - peak value:-

Period (T) : Time of a periodic waveform.

Cycle :



Frequency :- Number of cycles in 1 second.



$$\boxed{1\,Hz = 1\ cycle/s}$$

$f = 1/T$ and $T = 1/f$.

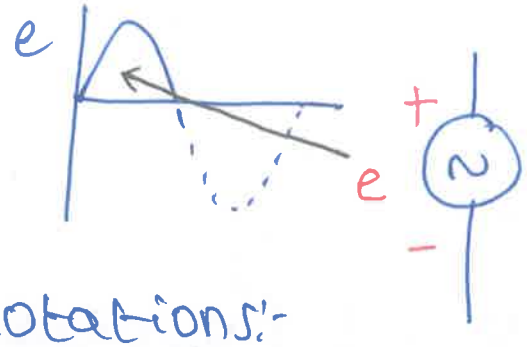
eg:- 60 find the period of a waveform with frequency (a) 50Hz. (b) 60Hz, c) 100Hz

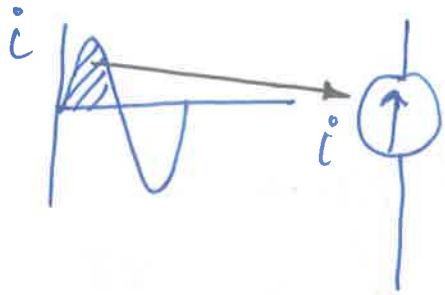
(a) $T = 1/f = 1/50 = 20ms.$

(b) $T = 1/f = 1/60 = 16.67ms$

(c) $T = 1/f = 1/100 = 10ms$

Polarities and direction :





notations:-

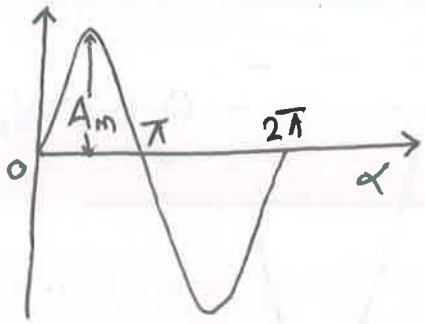
e, i, v for time varying (ac)

E, I, V for time invariant (dc)

Why sinusoidal and characteristics of sinusoidal?

Refer ppt.

Format of a Sinusoidal Voltage / Current :-



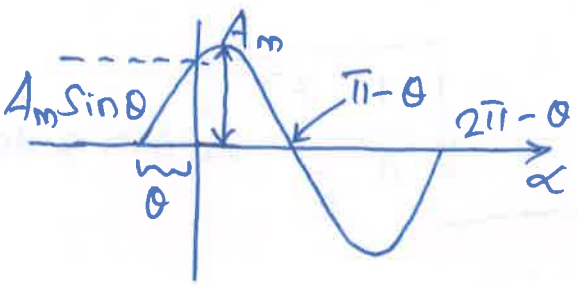
$A_m \sin\alpha$

$A_m \sin\omega t$

$i = I_m \sin\omega t = I_m \sin\alpha$

$e = E_m \sin\omega t = E_m \sin\alpha$

Phase relations :-

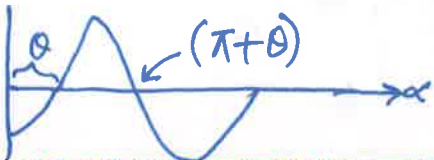


When wave is shifted to left or right to 0°

$A_m \sin(\omega t \pm \theta)$

BEFORE 0° :-

$\rightarrow A_m \sin(\omega t + \theta)$

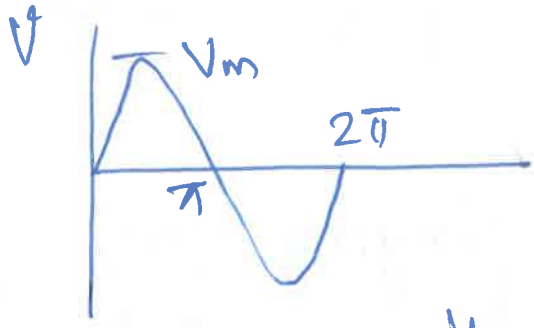


AFTER 0° :-

$\rightarrow A_m \sin(\omega t - \theta)$

Phase relations - avg. value - ppt

Average Value of a Sinusoidal Waveform:-

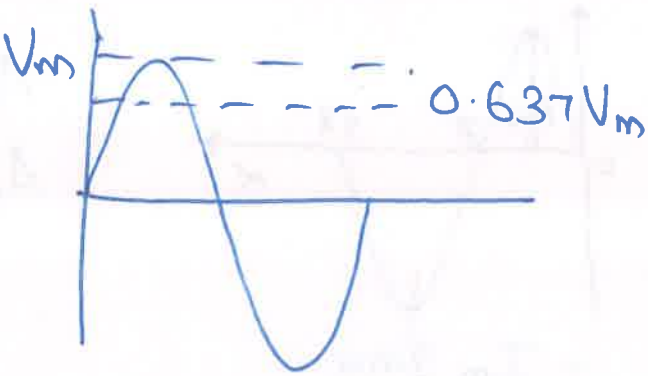


$$V_{av} = \frac{1}{\pi}\int_0^{\pi} V_m \sin\omega t \, d\omega t$$

$$V_{av} = \frac{V_m}{\pi}\left[-\cos\omega t\right]_0^{\pi}$$

$$= \frac{V_m}{\pi}\left[2\right] = \frac{2V_m}{\pi}$$

$$V_{av} = 0.637\, V_m$$



R.M.S Value of a Sinusoidal waveform:-

let $v(t) = V_m \sin\theta$

$$V_{RMS} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\, dt} = \sqrt{\frac{1}{\pi}\int_0^{\pi} V_m^2 \sin^2\theta \, d\theta}$$

$$= \sqrt{\frac{V_m^2}{\pi}\int_0^{\pi}\left[\frac{1-\cos 2\theta}{2}\right]_0^{\pi} d\theta}$$

$$= \sqrt{\frac{V_m^2}{\pi}\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi}} = \sqrt{\frac{V_m^2}{2\pi}\left[\pi\right]}$$

$$= \sqrt{\frac{V_m^2}{2}} = \frac{V_m}{\sqrt{2}}$$

$$I_{rms} = 0.707\, V_m.$$

Phasors:

Response of R, L, C to a Sinusoidal Voltage or Current:

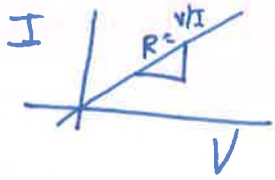
$\boxed{R}$

$$i = v/R = \frac{V_m \sin\omega t}{R} = \frac{V_m}{R}\sin\omega t = I_m \sin\omega t$$

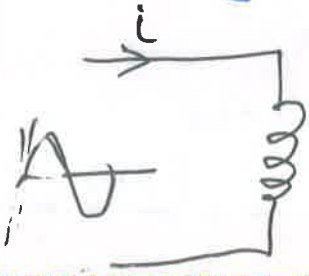
$$\boxed{I_m = V_m/R}$$

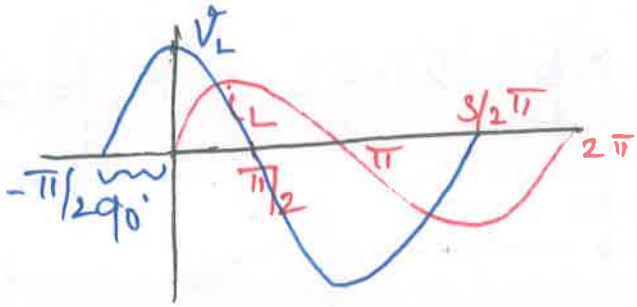
$$v = iR = I_m \sin\omega t \cdot R = (I_m R)\sin\omega t = V_m \sin\omega t$$

$$\boxed{V_m = I_m R}$$



$\boxed{L}$





$$v_L = L\frac{di}{dt} = L\frac{d}{dt}(I_m \sin\omega t)$$

$$= \omega L I_m \cos\omega t$$

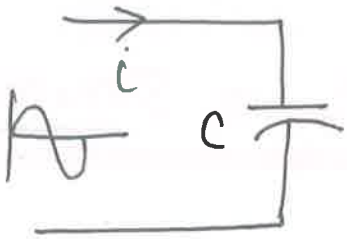
$$= X_L I_m \sin(\omega t + 90^\circ)$$

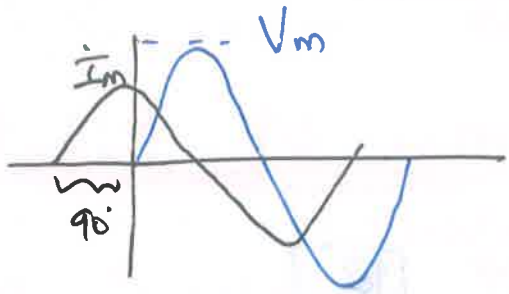
$$= V_m \sin(\omega t + 90^\circ) \qquad [V_m = \omega L I_m]$$

V leads I by 90°.

$$X_L = \frac{V_m}{I_m}$$

C





$$i_C = C\frac{dv_C}{dt} = C\frac{d V_m \sin\omega t}{dt}$$

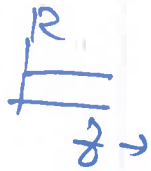
$$i_C = \omega C V_m \cos\omega t.$$

$$i_C = \frac{V_m}{X_C}\sin(\omega t + 90^\circ)$$

$$\boxed{\text{i leads V by } 90^\circ.}$$

Ideal Response:-

R :- Indep. of frequency



L : $X_L = 2\pi f L$ $\quad X_L\big|_{f=0} = 0$; $X_L\big|_{f=\infty} = \infty$

C: $X_C = \frac{1}{2\pi f C}$.

Complex Numbers:-

Complex numbers may be represented in either rectangular form or polar form.

a) Rectangular form:- $C = x + iy$

b) polar form :- $C = Z\angle\theta^\circ$

Conversion:-

Rec. → polar $\quad Z = \sqrt{x^2+y^2}$.

polar → Rect. $\quad \theta = \tan^{-1}(Y/X)$.

$X = Z\cos\theta$.

$Y = Z\sin\theta$.

Mathematical operations:-

(a) addition:- (b) subtraction.

If $C_1 = x_1 \pm jy_1$

$C_2 = x_2 \pm jy_2$

$C = C_1 + C_2 = (x_1 \pm x_2) + j(y_1 \pm y_2)$.

eg:62 add $C_1 = 1+2j$, $C_2 = 3+5j$

$C = 4+7j$

eg: 63

$C_1 = 2 - j3$, $C_2 = -5 + 5j$

$C = -3 + 3j$

~~(b) Subtraction~~ :- (c) Multiplication (In rect)

$$C_1 = x_1 + j y_1$$
$$C_2 = x_2 + j y_2$$
$$C_1 C_2 \quad x_1 x_2 + j y_2 x_1$$
$$j x_2 x_1 + j^2 y_1 y_2$$
$$(x_1 x_2 - y_1 y_2) + j(x_2 y_1 + x_1 y_2)$$

In polar:

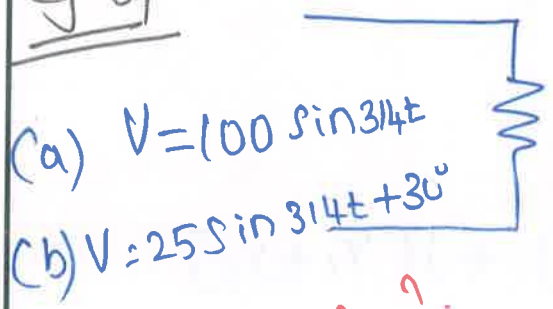
$$C_1 = Z_1 \angle \theta_1 \quad C_2 = Z_2 \angle \theta_2 .$$
$$C_1 C_2 = Z_1 Z_2 \angle \theta_1 + \theta_2$$

d) Division:

$$C_1 / C_2 = Z_1 / Z_2 \angle \theta_1 - \theta_2 .$$

Problems:

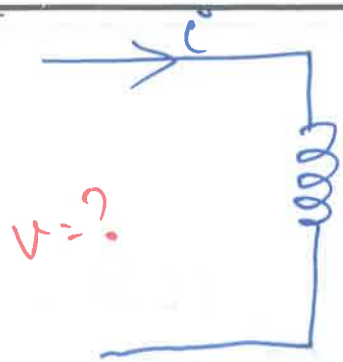
eg: 64



(a) $i = 10 \sin 314t$.

(b) $i = 2.5 \sin(314 + 30)$

eg: 65



(a) $i = 10 \sin 314t$.

(b) $i = 7 \sin(314t - 70^\circ)$.

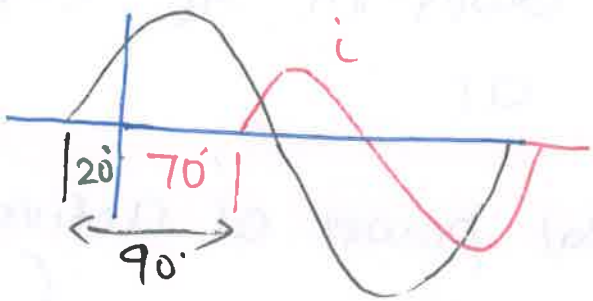
$X_L = 2\pi f L = 2\pi \times 50 \times 0.1$

$= 314 \times 0.1 = 31.4\,\Omega$.

(a) $v = i X_L = 314 \sin(314t + 90^\circ)$.

(b) $v = 7 \times 31.4 \sin(314t - 70 + 90^\circ)$

$= 219.8 \sin(314t + 20^\circ)$.



eg: 66 Convert R → P

(a) $3 + j4 \Rightarrow Z = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$

$\theta = \tan^{-1}(4/3) = 53.13^\circ$

$5 \angle 53.13^\circ$

P – R

(b) $10 \angle 45^\circ$

$x = 10 \cos 45^\circ = 7.07$

$y = 10 \sin 45^\circ = 7.07$

$c = 7.07 + j7.07$.

Complex Conjugate:

$C = 2 + j3$ $\qquad$ $C = 2\angle 30^\circ$

$C^* = 2 - j3$ $\qquad$ $C^* = 2\angle -30^\circ$

Need for phasor:-

Refer ppt.

Power in AC Circuit:-

Power in AC Circuit can be classified as

(i) Real power or active power or true power (P)

(ii) Reactive power (Q)

(iii) Apparant power (S)

P:

$$P_{av} = V_{rms} I_{rms} \underbrace{\cos\theta}_{\text{power factor.}}$$

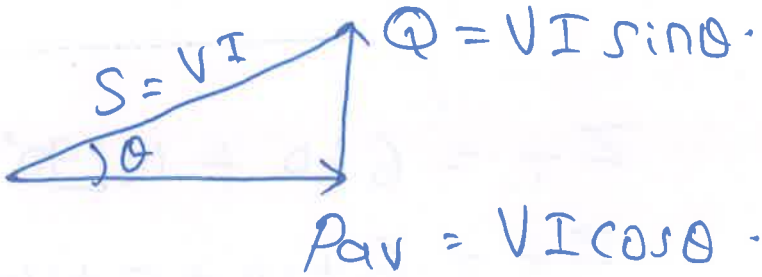
$$= \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos\theta$$

$$P_{av} = \frac{V_m I_m}{2} \cos\theta \quad (W)$$

Q: $Q = V_{rms} I_{rms} \sin\theta$ (VAR)

S: $S = V_{rms} I_{rms}$

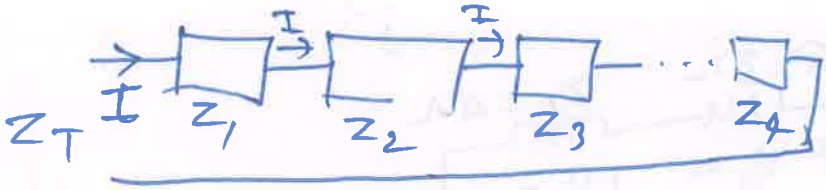
Power triangle:



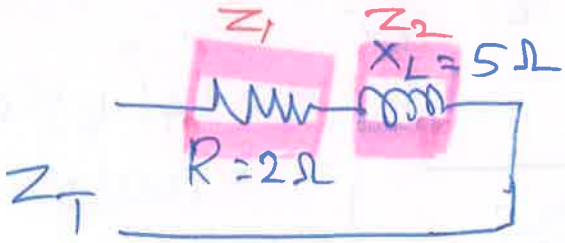
Series Configuration in AC circuit:-

Properties of ac circuits and dc circuits are same for series configuration.

$$Z_T = Z_1 + Z_2 + \dots + Z_n.$$



eg: 72



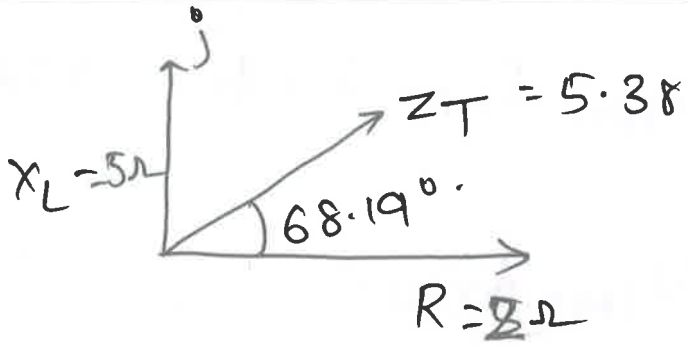
$$Z_T = Z_1 + Z_2$$

$$= 2\angle 0^\circ + 5\angle 90^\circ$$

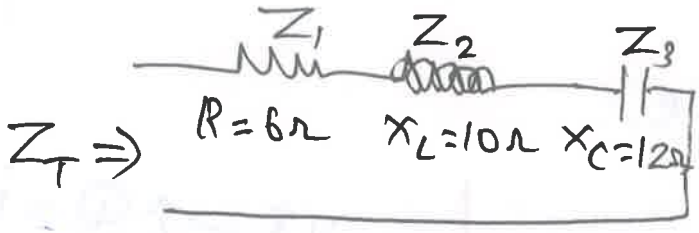
$$= 2 + j0 + 0 + j5$$

$$Z_T = 2 + j5$$

$$= 5.38\angle 68.19 \dots$$



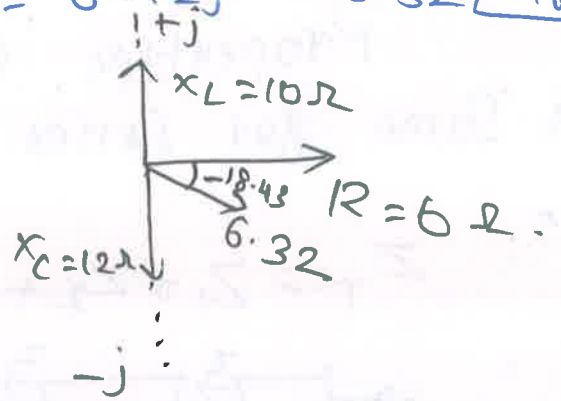
eg: 73



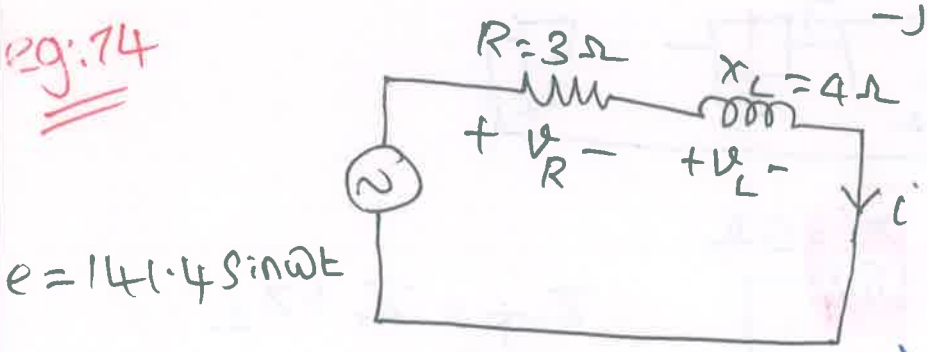
$$Z_T = 6\angle 0° + 10\angle 90° + 12\angle -90°$$

$$Z_T = 6 + j0 + 0 + j10 + 0 - 12j$$

$$= 6 + j(10-12) = 6 - 12j = 6.32\angle -18.43°$$

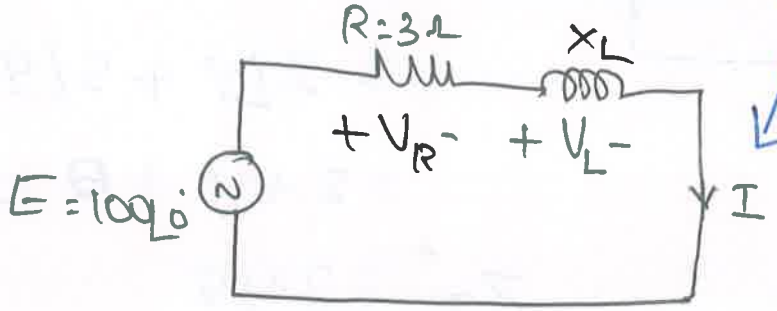


eg: 74



$E_m = \sqrt{2} E_{rms}$

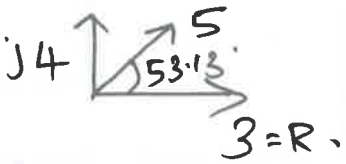
$$e = 141.4 \sin \omega t$$
$$= 100\angle 0°$$



Phasor notation
(Time Indep. Ckt)

$$Z_T = Z_1 + Z_2 = 3\angle 0^\circ + 4\angle 90^\circ = 3 + j4\,\Omega$$

$$= 5\angle 53.13^\circ$$



$$I = \frac{E}{Z_T} = \frac{100\angle 0^\circ}{5\angle 53.13^\circ} = 20\angle -53.13^\circ$$

V_R: $V_R = IR = (20\angle -53.13)(3\angle 0^\circ)$

$$= 60\angle -53.13$$

$$V_L = I X_L = (20\angle -53.13)(4\angle 90^\circ)$$

$$= 80\angle 36.87^\circ$$

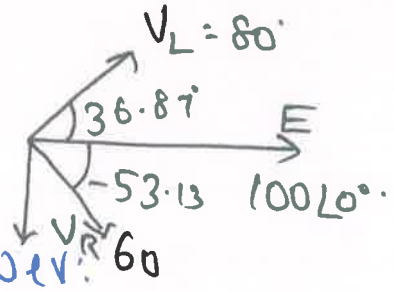
Verifying KVL

$$E = V_R + V_L$$

$$100\angle 0^\circ = 60\angle -53.13 + 80\angle 36.87$$

$$= 36 - j48 + 64 + j48$$

$$= 100V + j0 = 100\angle 0^\circ$$



Power:

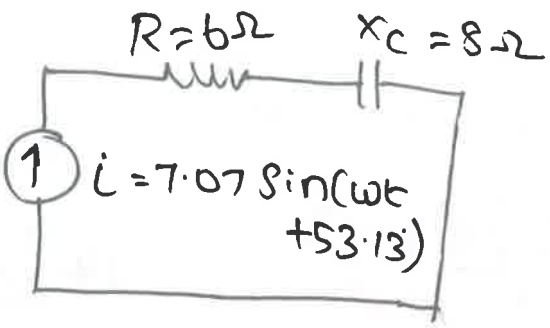
$$P_T = EI\cos\theta_T = 100 \times 20 \cos\theta_T \quad (53.13^\circ)$$

$$= 1200W$$

Power factor, $F_P = \cos\theta_T = \cos 53.13$

$$F_P = \frac{R}{Z_T} = \frac{3}{5} = 0.6 \text{ lagging}$$

eg: 75



e, v_R, v_C

Phasor notation:

$$i = 7.07 \sin(\omega t + 53.13^\circ)$$

$$\bar{I} = 5A \angle 53.13^\circ$$

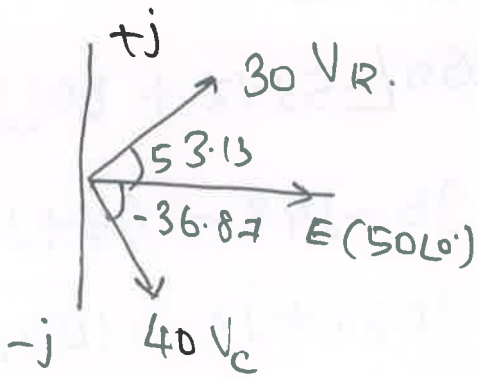
$$Z_T = 6\angle 0^\circ + 8\angle -90^\circ = 6 + j0 + 0 - j8 = 6 - j8\,\Omega$$

$$Z_T = 10\Omega \angle -53.13\,\Omega$$

$$E = IZ_T = 50\angle 0^\circ \quad \{5\angle 53.13 \times 10\angle -53.13\}$$

$$V_R = IR = 5\angle 53.13 \times 6\angle 0^\circ = 30\angle 53.13^\circ$$

$$V_C = IX_C = 5\angle 53.13 \times 8\angle -90^\circ = 40\angle -36.87^\circ$$



In time domain.

$$e = \sqrt{2}(50)\sin\omega t = 70.70\sin\omega t.$$

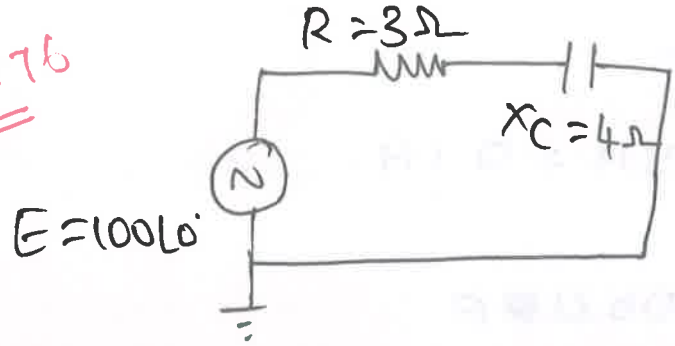
$$v_R = \sqrt{2}(30)\sin\omega t + 53.13 = 42.42\sin(\omega t + 53.13^\circ)$$

$$v_C = \sqrt{2}(40)\sin(\omega t - 36.87) = 56.56\sin(\omega t - 36.87)$$

Voltage divider rule (ac):-

$$V_x = E \frac{Z_x}{Z_T}.$$

eg: 76



$V_C, V_R, ?$

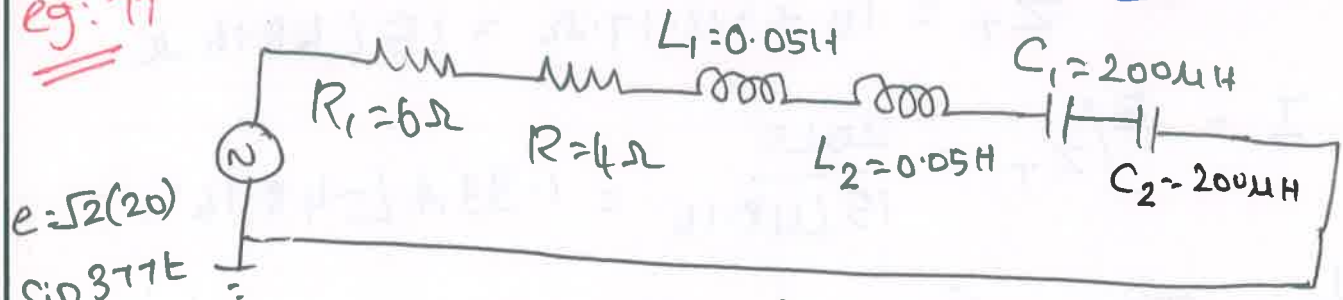
$$V_C = (100\angle 0^\circ)\frac{4\angle -90^\circ}{3\angle 0^\circ + 4\angle -90^\circ}$$

$$V_C = \frac{400\angle -90^\circ}{3 - j4} = \frac{400\angle -90^\circ}{5\angle -53.13} = 80\angle -36.87^\circ.$$

$$V_R = (100\angle 0^\circ)\frac{3\angle 0^\circ}{3\angle 0 + 4\angle -90^\circ} = \frac{300\angle 0^\circ}{5\angle -53.13}$$

$$= 60\angle 53.13^\circ$$

eg: 77



let f = 60Hz

a) find I, V_R, V_L, V_C in phasor form.

b) F_Q

c) P_{av} of the circuit.

Phasor representation:-

Combining the elements and reactances,

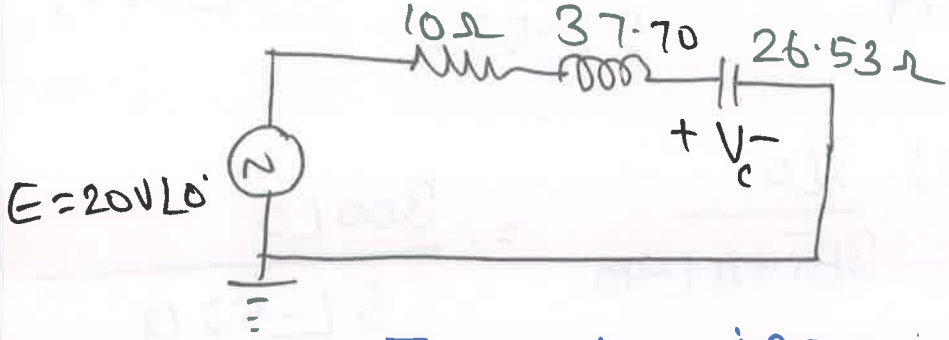
$R_T = 6\Omega + 4\Omega = 10\Omega$

$L_T = 0.05H + 0.05H = 0.1H.$

$C_T = \frac{200\mu F}{2} = 100\mu F$

$X_L = \omega L = 377 \times 0.1 = 37.70\Omega.$

$X_C = 1/\omega C = \frac{1}{377 \times 100\mu F} = 26.53\Omega.$



$Z_T = 10 + j37.70 - j26.53$

$Z_T = 10 + j11.17\Omega = 15\angle 48.16\Omega.$

$I = E/Z_T = \frac{20\angle 0^\circ}{15\angle 48.16} = 1.33A\angle -48.16^\circ.$

$V_R = IZ_R = (1.33\angle -48.16)(10\angle 0^\circ) = 13.3\angle -48.16\ V$

$V_L = IZ_L = (1.33\angle -48.16)(37.70\angle +90^\circ) = 50.14\angle 41.84^\circ$

$V_C = IZ_C = 35.28\angle -138.16$

$F_p = \cos\theta = \cos 48.16 = 0.667$ lag.

$P_{avT} = EI\cos\theta = 20 \times 1.33 \times 0.667 = 17.74\,W.$

Parallel AC Circuits:-

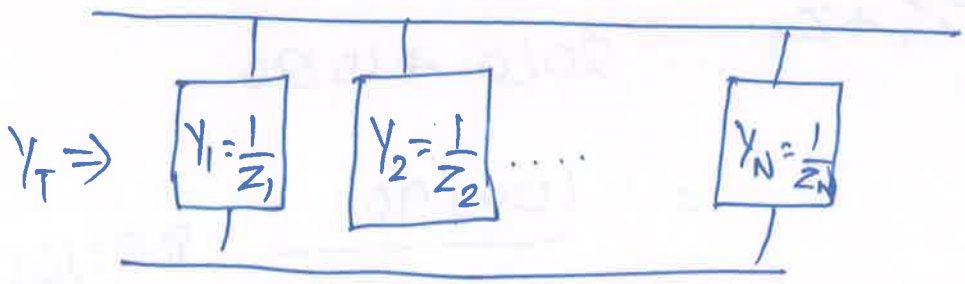
Admittance & Susceptance.

In dc circuits, conductance ($\cancel{1/G}$ G) was referred to $1/R$.

In ac circuits, ~~impedance~~ admittance (Y) ~~w~~ is referred to $1/Z$

$$Y_T = Y_1 + Y_2 + Y_3 + \cdots + Y_N$$

$$\frac{1}{Z_T} = \frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3} + \cdots + \frac{1}{Z_N}$$



Admittance becomes Conductance if $Z = R$

$Y \longrightarrow G$, if $Z = R$.

$Y \longrightarrow$ Susceptance ($1/X$) if $Z = X_L \angle 90^\circ$

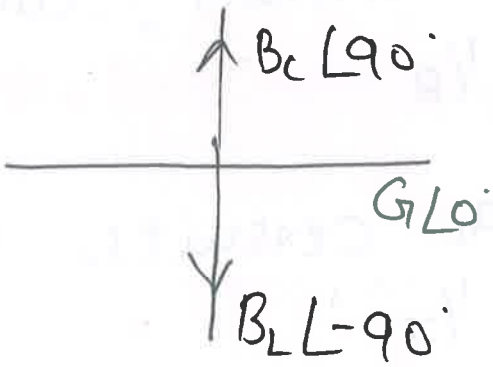
or

$Z = X_C \angle -90^\circ$

$$Y_R = \frac{1}{Z_R} = \frac{1}{R\angle 0^\circ} = G\angle 0^\circ$$

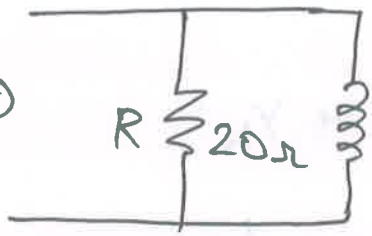
$$Y_L = \frac{1}{Z_L} = \frac{1}{X_L\angle 90^\circ} = B_L\angle -90^\circ$$

$$Y_C = \frac{1}{Z_C} = \frac{1}{X_C\angle -90^\circ} = B_C\angle 90^\circ$$



eg: 78

$Y_T \Rightarrow$ Z_T



Draw also imp. & adm. diagram

$$Z_T = \frac{Z_1 Z_2}{Z_1 + Z_2} = \frac{(20\angle 0^\circ)(10\angle 90^\circ)}{20\angle 0^\circ + 10\angle 90^\circ}$$

$$= \frac{100\angle 90^\circ}{20 + j10} = 8.93\angle 63.43^\circ$$

$$Y_T = Y_1 + Y_2$$

$$= \frac{1}{R} + \frac{1}{X_L} = \frac{1}{20\angle 0^\circ} + \frac{1}{10\angle 90^\circ} = 0.05\angle 0^\circ + 0.1\angle -90^\circ$$

$Y_R + Y_L$

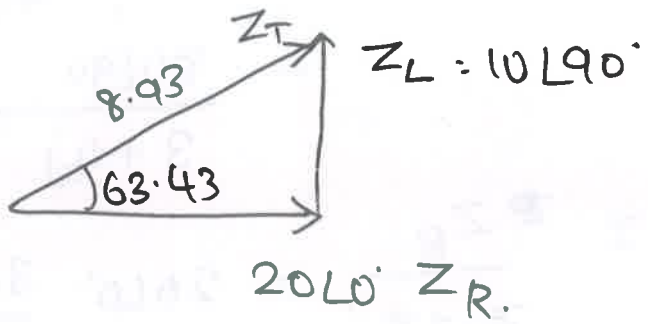
$$= 0.05 - j0.1\,S$$

or

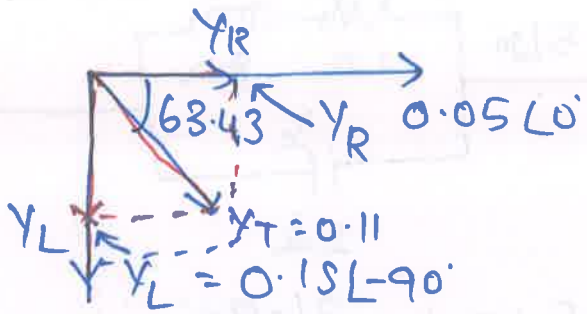
$$Y_T = \frac{1}{Z_T} = \frac{1}{8.93\angle 63.43} = 0.11\angle -63.43$$

$$Y_T = 0.05 - 0.01j \text{ S.}$$

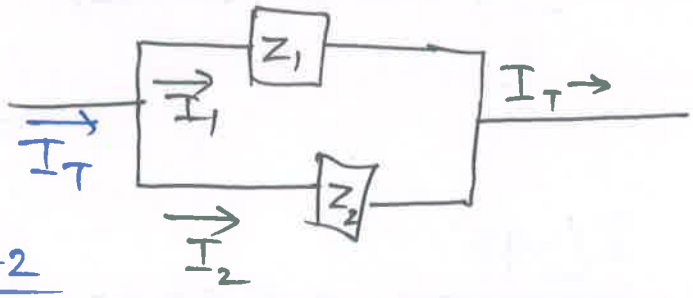
Imp. diagram:



Adm. diagram:

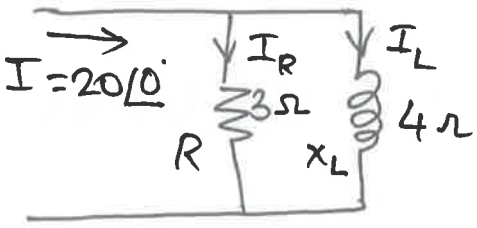


Current divider rule:-



$$I_1 = I_T \frac{Z_2}{Z_1 + Z_2}$$

eg: 79



$I_R, I_L = ?$

$$I_R = I \frac{Z_L}{Z_L + Z_R}$$

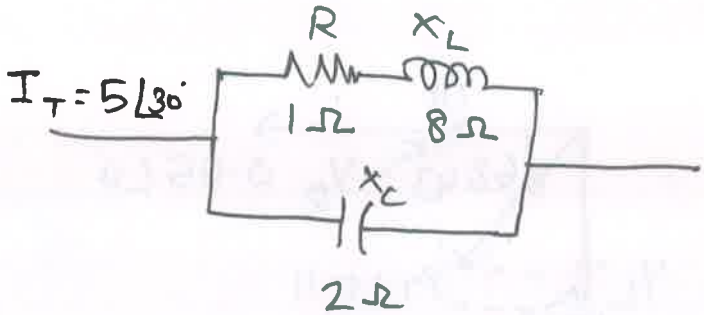
$$= 20\angle 0^\circ \frac{4\angle 90^\circ}{4\angle 90^\circ + 3\angle 0^\circ}$$

$$= \frac{80\angle 90^\circ}{3+4j} = 16\angle 36.87^\circ$$

$$I_L = I \frac{Z_R}{Z_R + Z_L} = 20\angle 0^\circ \frac{3\angle 0^\circ}{3\angle 0^\circ + 4\angle 90^\circ} = \frac{60\angle 0^\circ}{3+4j}$$

$$= 12\angle -53.13^\circ$$

eg: 80



$I_{RL}, I_c = ?$

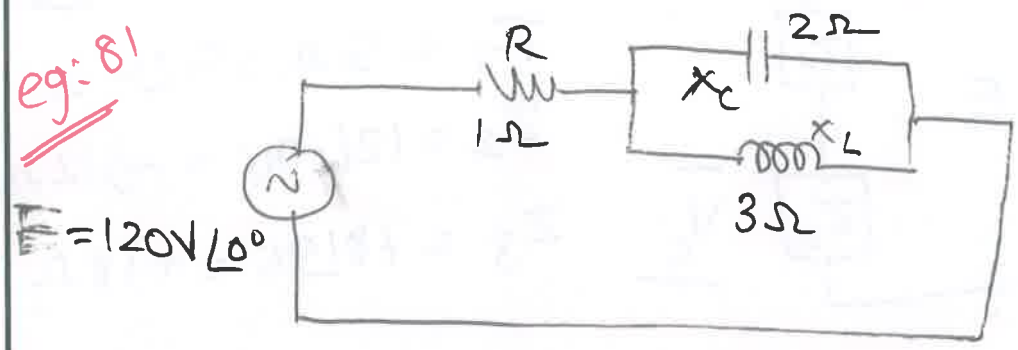
$$I_{R-L} = 5\angle 30^\circ \frac{2\angle -90^\circ}{(2\angle -90^\circ) + (1+8j)} = \frac{10\angle -60^\circ}{1+8j-2j}$$

$$= \frac{10\angle -60}{6.083\angle 80.54} = 1.64\angle -140.5$$

$$I_c = 5\angle 30 \frac{2\angle -90^\circ}{1+8j-2j} = 6.63\angle 32.33^\circ$$

Series - parallel ac circuits:

eg: 81



$I_s = ?$

$Z_1 = 1 + j0 = 1\angle 0^\circ$

$Z_2 = Z_C \| Z_L$

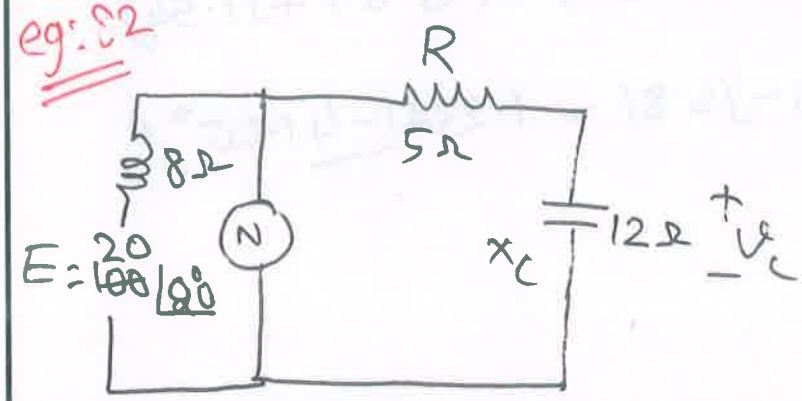
$= \dfrac{(2\angle -90^\circ)(3\angle 90^\circ)}{-j2 + j3}$

$= \dfrac{6\angle 0^\circ}{1\angle 90^\circ} = 6\angle -90^\circ$

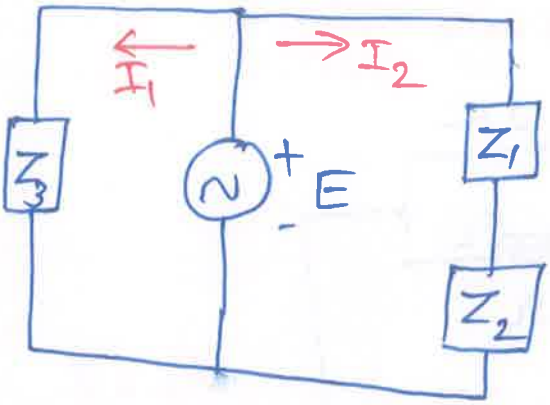
$Z_T = Z_1 + Z_2 = (1 + j0) + (0 + 6j) = 1 + 6j = 6.08\angle -80.5^\circ$

$I_s = \dfrac{E}{Z_T} = \dfrac{120\angle 0^\circ}{6.08\angle -80.5^\circ} = 19.74\angle 80.54^\circ$

eg: 82



$V_C = ?, \; I_s = ?$



$\bar{V}_c$

$Z_1 = 5\Omega = 5\angle 0^\circ\ \Omega$

$Z_2 = 12\angle -90^\circ = -j12\Omega$

$Z_3 = +8\angle 90^\circ = +j8\Omega.$

$$V_c = E\frac{Z_2}{Z_1+Z_2} = \frac{(12\angle -90^\circ)(20\angle 20^\circ)}{5\Omega - j12\Omega}$$

$$V_c = \frac{240V\angle -70^\circ}{13\angle -67.38^\circ} = 18.46\angle -2.62^\circ V$$

$$I_1 = \frac{E}{Z_3} = \frac{20\angle 20^\circ}{8\angle 90^\circ} = 2.5\angle -70^\circ$$

$$I_2 = \frac{E}{Z_1+Z_2} = \frac{20\angle 20^\circ}{5-12j} = \frac{20\angle 20^\circ}{13\angle -67.38^\circ} = 1.54\angle 87.38^\circ$$

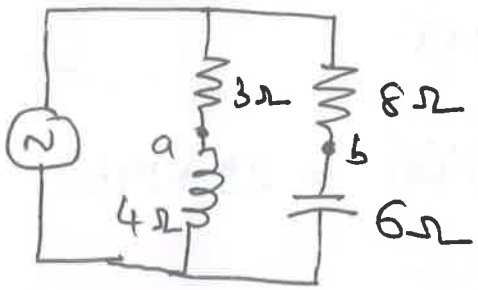
$$I_s = I_1 + I_2 = 2.5\angle -70^\circ + 1.54\angle 87.38^\circ$$

$$= (0.86 - j2.35) + (0.07 + j1.54)$$

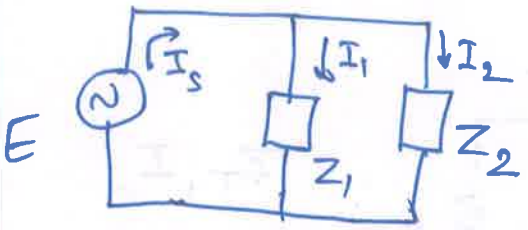
$$I_s = 0.93 - j0.81 = 1.23A\angle -41.05^\circ$$

eg.83

$E = 100\angle 0^\circ$



$I_s = ?, \; V_{ab} = ?$



$$Z_1 = 3+j4 = 5\angle 53.13^\circ$$

$$Z_2 = 8-j6 = 10\angle -36.87^\circ$$

$$Z_T = \frac{Z_1 Z_2}{Z_1+Z_2} = \frac{(5\angle 53.13)(10\angle -36.87)}{(3+j4)+(8-j6)}$$

$$= \frac{50\angle 16.26^\circ}{11-j2} = \frac{50\Omega\angle 16.26^\circ}{11.18\angle -10.30^\circ}$$

$$= 4.472\angle 26.56^\circ$$

$$I_s = \frac{E}{Z_T} = \frac{100\angle 0^\circ}{4.47\angle 26.56} = 22.36A\angle -26.56^\circ$$

$$I_1 = \frac{E}{Z_1} = \frac{100\angle 0^\circ}{5\angle 53.13} = 20A\angle -53.13^\circ$$

$$I_2 = \frac{E}{Z_2} = \frac{100V\angle 0^\circ}{10\angle -36.87^\circ} = 10A\angle 36.87^\circ$$

$$V_{ab} + V_{R_1} - V_{R_2} = 0$$

$$V_{ab} = V_{R_2} - V_{R_1} = I_2 R_2 - I_1 R_1$$

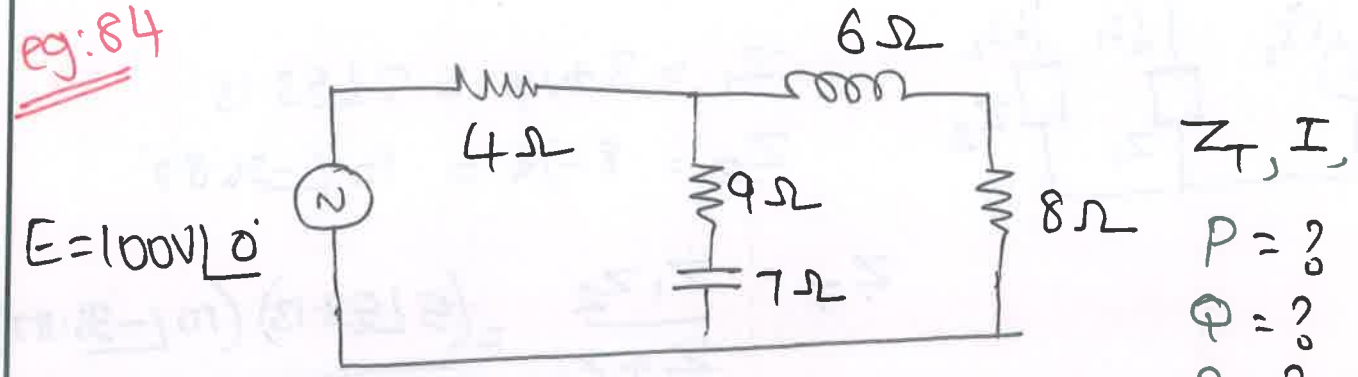
$$= (10\angle 36.87^\circ)8\Omega - (20\angle -53.13)(3\Omega)$$

$= 80\angle 36.87^\circ - 60\angle -53.13^\circ$

$= (64 + j48) - (36 - j48) = 28 + j96.$

$V_{ab} = 100V\angle 73.74^\circ.$

eg: 84

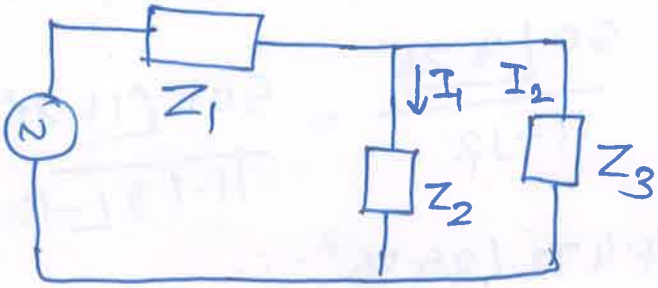


$Z_T, I,$

$P = ?$

$Q = ?$

$S = ?$



$Z_1 = 4 + j0 = 4\angle 0^\circ$

$Z_2 = 9 - 7j = 11.4\angle -37.87\Omega$

$Z_3 = 8 + j6\Omega = 10\angle 36.87\Omega$

$Z_T = Z_1 + (Z_2 \| Z_3).$

$= Z_1 + \dfrac{Z_2 Z_3}{Z_2 + Z_3} = (4 + j0) + \left\{ \dfrac{(9 - j7)(8 + j6)}{9 - j7 + 8 + j6} \right\}$

$= (4 + j0) + \dfrac{114 - 2j}{17 - j}$

$= 10.69 + 0.28j$

$= 10.69\angle 1.5^\circ\ \Omega$

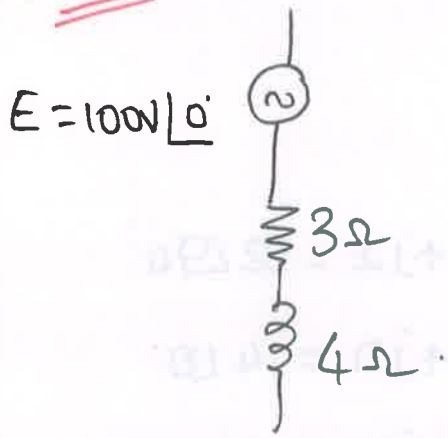
$$I = \frac{E}{Z_T} = \frac{100\angle 0^\circ}{10.69\angle 1.5^\circ} = 9.35\angle -1.5^\circ \text{ A.}$$

$$P = EI\cos\theta = 100 \times 9.35 \cos 1.5^\circ = 934.7\text{ W.}$$

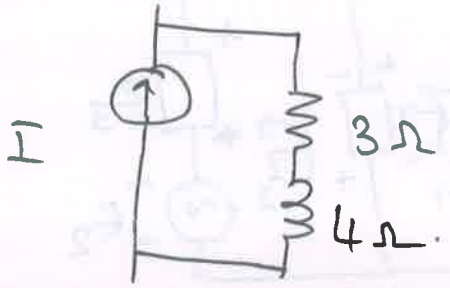
$$Q = EI\sin\theta = 24.475 \text{ VAR.}$$

$$S = EI = 935 \text{ VA.}$$

eg: 85



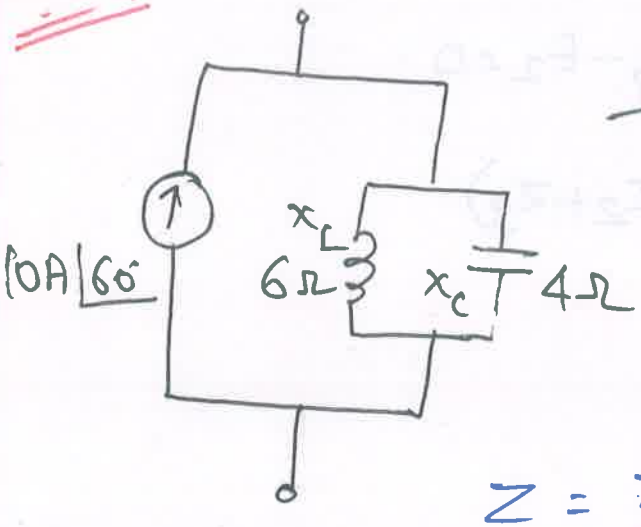
Convert to current source:



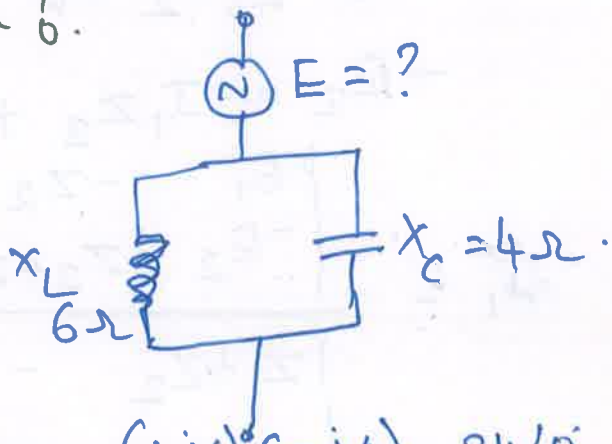
$$Z = 3 + j4\,\Omega.$$

$$I = E/Z = \frac{100}{5\angle 53.13} = 20\angle -53.13^\circ$$

eg: 86



V_{source} ?



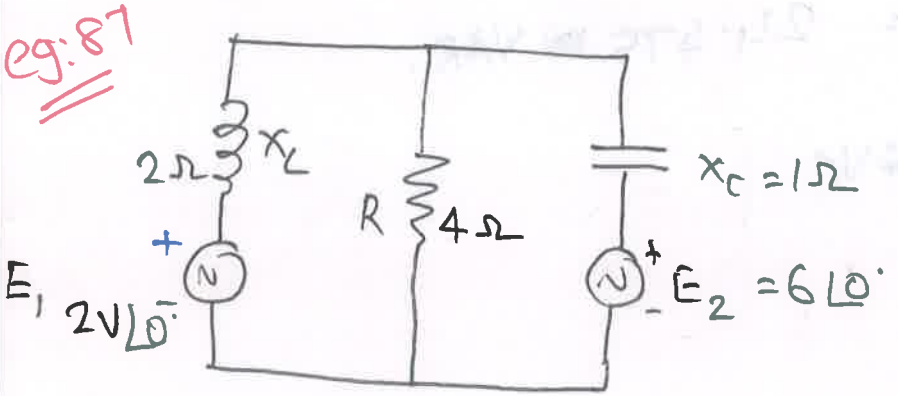
$$Z = \frac{Z_C Z_L}{Z_C + Z_L} = \frac{(+j6)(-j4)}{+j2} = \frac{24\angle 0^\circ}{2\angle 90} = 12\angle -90^\circ$$

$$E = IZ = (10A\angle 60^\circ)(12\Omega\angle -90^\circ)$$

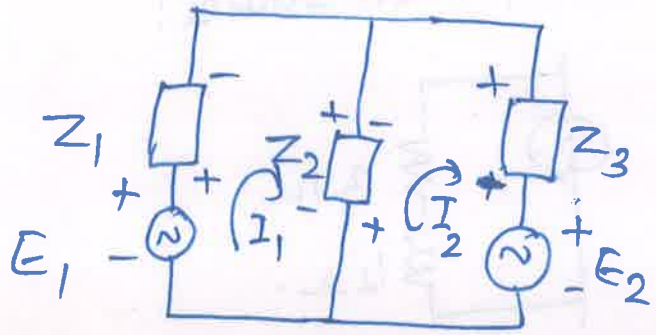
$$E = 120V\angle -30^\circ$$

Mesh analysis (AC)

eg:87



$I_1 = ?$



$Z_1 = 0 + j2 = 2\angle 90^\circ$

$Z_2 = 4 + j0 = 4\angle 0^\circ$

$Z_3 = 0 - j1 = 1\angle -90^\circ$

L1:

$$E_1 - I_1 Z_1 - Z_2(I_1 - I_2) = 0.$$

$$E_1 = I_1(Z_1 + Z_2) - I_2 Z_2$$

L2:

$$-Z_2(I_2 - I_1) - I_2 Z_3 - E_2 = 0.$$

$$-E_2 = -I_1 Z_2 + I_2(Z_2 + Z_3)$$

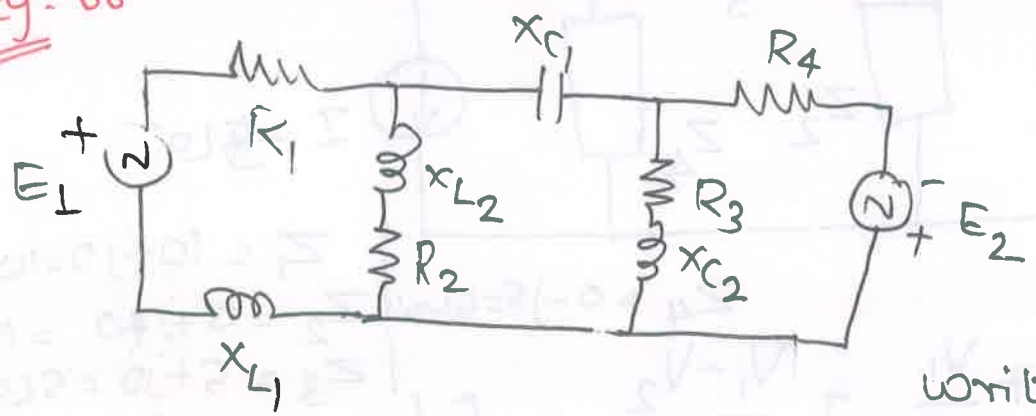
$$I_1 = \frac{\begin{vmatrix} E_1 & -Z_2 \\ -E_2 & Z_2 + Z_3 \end{vmatrix}}{\begin{vmatrix} Z_1 + Z_2 & -Z_2 \\ -Z_2 & Z_2 + Z_3 \end{vmatrix}}$$

$$I_1 = \frac{E_1(Z_2+Z_3) - E_2 Z_2}{(Z_1+Z_2)(Z_2+Z_3) - Z_2^2}$$

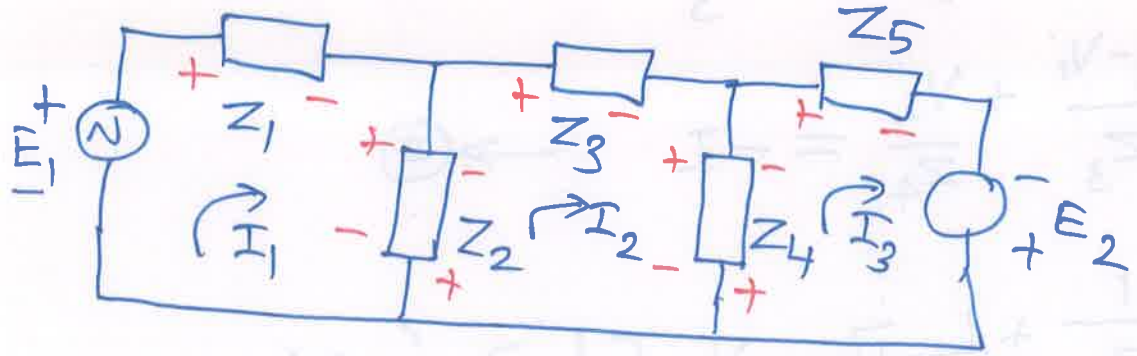
$$= \frac{(2+j0)(4-j1) - (6\times 4)}{(4+2j)(4-j1) - 16} = \frac{-16+2j}{2+4i}$$

$$= -1.2+3.4j = 3.6 \angle 109.4^\circ$$

eg: 88



write mesh equations.



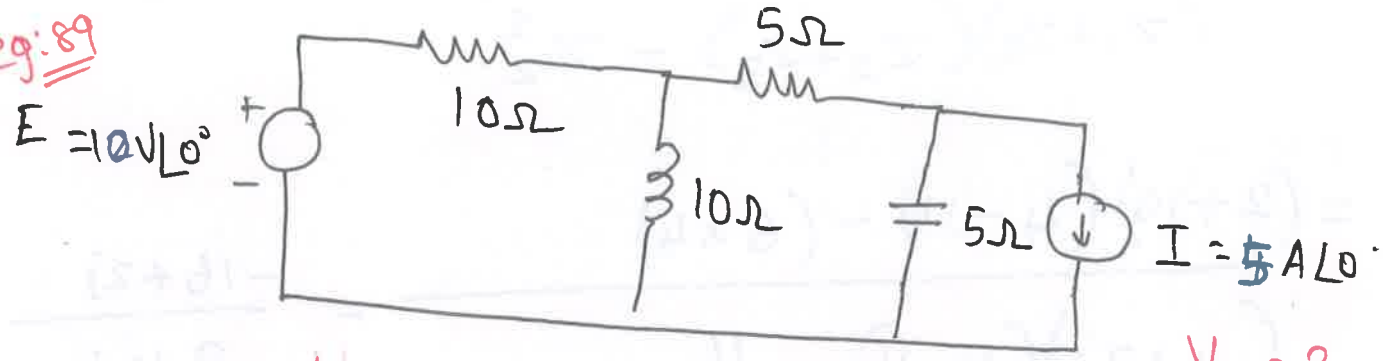
$$I_1(Z_1+Z_2) - I_2(Z_2) - I_3(0) = E_1$$

$$-I_1(Z_2) + I_2(Z_2+Z_3) - I_3(Z_4) = 0$$

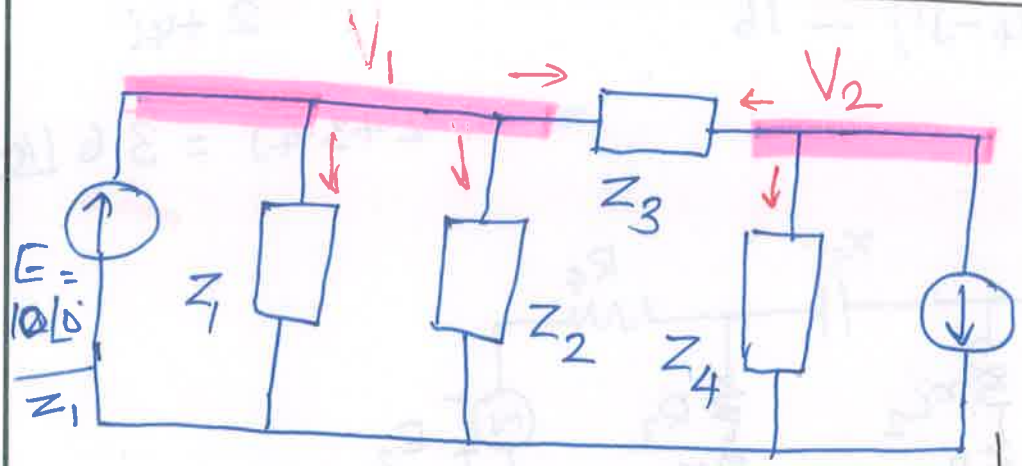
$$-I_1(0) - I_2(Z_4) + I_3(Z_4+Z_5) = -E_2$$

Nodal analysis (ac) :-

eg: 89



$V_L = ?$



$Z_4 = 0 - j5 = 5\angle -90°$

$Z_1 = 10 + j0 = 10\angle 0°$

$Z_2 = 0 + j10 = 10\angle 90°$

$Z_3 = 5 + j0 = 5\angle 0°$

$$\frac{V_1}{Z_1} + \frac{V_1}{Z_2} + \frac{V_1 - V_2}{Z_3} = E/Z_1 \longrightarrow (1)$$

$$\frac{V_2 - V_1}{Z_3} + \frac{V_2}{Z_4} = -I \longrightarrow (2)$$

$$V_1\left[\frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3}\right] - V_2\left[\frac{1}{Z_3}\right] = E/Z_1$$

$$-V_1\left[\frac{1}{Z_3}\right] + V_2\left[\frac{1}{Z_3} + \frac{1}{Z_4}\right] = -I$$

$$V_1(0.3 - \overset{0.1}{\cancel{10}}j) - V_2(0.2) = 1\,V$$

$$-V_2(0.2) + V_2(0.2 + 0.2j) = -5$$

$$V_1 = \frac{\begin{vmatrix} E/Z_1 & -1/Z_3 \\ -I & \frac{1}{Z_3}+\frac{1}{Z_4} \end{vmatrix}}{\begin{vmatrix} \frac{1}{Z_1}+\frac{1}{Z_2}+\frac{1}{Z_3} & -\frac{1}{Z_3} \\ -1/Z_3 & \frac{1}{Z_3}+\frac{1}{Z_4} \end{vmatrix}} = \frac{\begin{vmatrix} 1 & -0.2 \\ -5 & 0.2+0.2j \end{vmatrix}}{\begin{vmatrix} 0.3-\cancel{10}\,0.1j & -0.2 \\ -0.2 & 0.2+0.2j \end{vmatrix}}$$

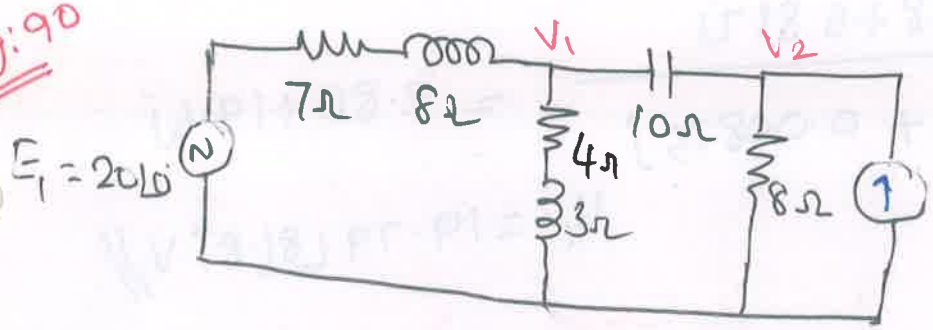
98

$$V_1 = \frac{1(0.2+0.2j) - 5(0.2)}{(0.3-\cancel{10}\,0.1j)(0.2+0.2j) - (0.2)(0.2)}$$

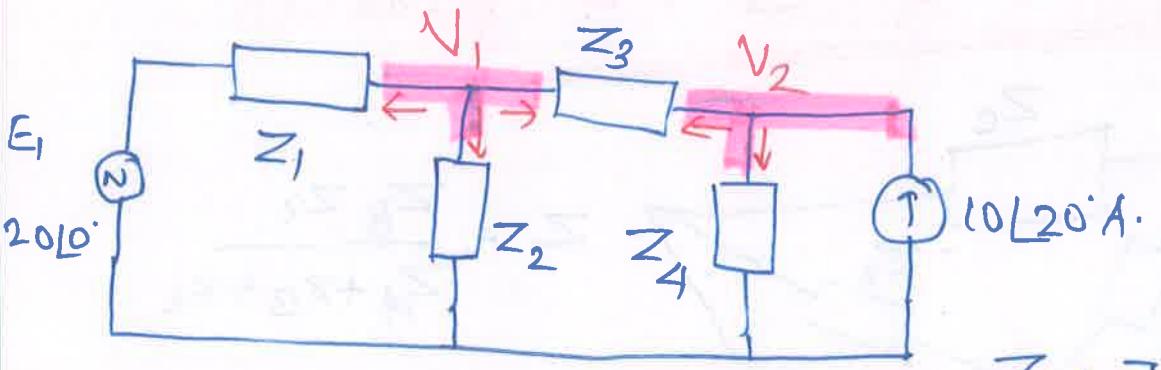
$$V_1 = \frac{-0.8+0.2j}{\cancel{2.06-1.94j-0.04}\;\; 0.04+0.04j} = -0.255-0.146j\ V.$$

$$V_1 \doteq \cancel{0.294\angle -150^\circ V.} = V_L = -7.5+12.5j$$

eg: 90



V_1, ~~V_2~~ ?



$Z_1 = 7+8j =$

$Z_2 = 4+3j$

$Z_3 = 0-10j$

$Z_4 = 8+j0$

$$V_1\left[\frac{1}{Z_1}+\frac{1}{Z_2}+\frac{1}{Z_3}\right]-V_2\left[1/Z_3\right]=E_1/Z_1$$

$$-V_1\left[1/Z_3\right]+V_2\left[1/Z_3+1/Z_4\right]=+I_1$$

$$V_1(0.22-0.09j)-V_2(0.1j)=1.23-1.41j$$

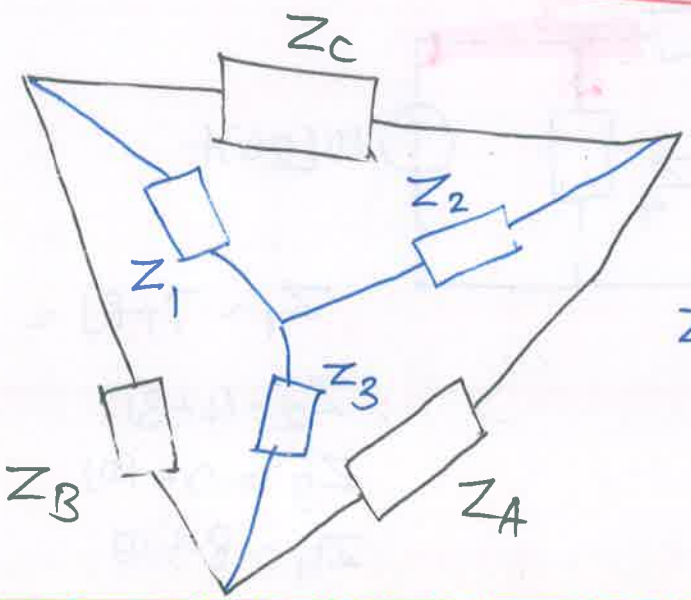
$$-V_1(0.1j)+V_2(0.125+0.1j)=+9.39+3.42j$$

$$V_1=\frac{\begin{vmatrix}1.23-1.41j & -0.1j\\ +9.39+3.42j & (0.125+0.1j)\end{vmatrix}}{\begin{vmatrix}0.22-0.09j & -0.1j\\ -0.1j & 0.125+0.1j\end{vmatrix}}$$

$$V_1=\frac{-0.048+0.817j}{0.044+0.00875j}=2.80+19.6j$$

$$V_1=19.79\angle 81.8^\circ\ V$$

Δ-λ and λ-Δ transformation:-



$$Z_1=\frac{Z_B Z_C}{Z_A+Z_B+Z_C}$$

$$Z_2=\frac{Z_A Z_C}{Z_A+Z_B+Z_C}$$

$$Z_3 = \frac{Z_A Z_B}{Z_A + Z_B + Z_C}$$

$$Z_A = \frac{Z_1 Z_2 + Z_1 Z_3 + Z_2 Z_3}{Z_1} = \frac{\Delta}{Z_1}$$

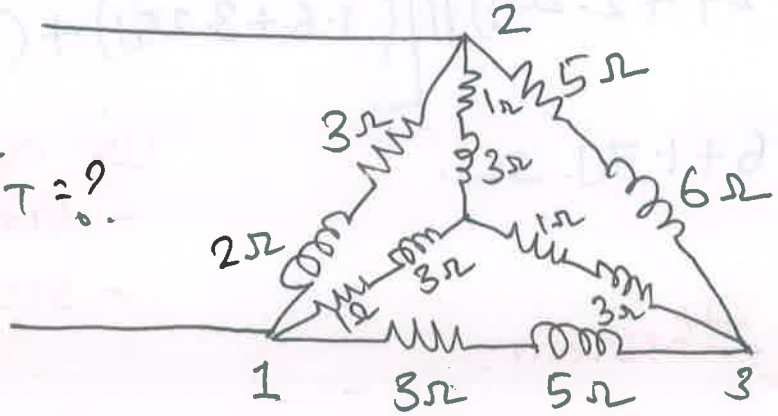
$$Z_B = \frac{\Delta}{Z_2} \quad ; \quad Z_C = \frac{\Delta}{Z_3}$$

If $Z_1 = Z_2 = Z_3$;

$$Z_\Delta = 3 Z_\lambda$$

eg: 91

$Z_T = ?$

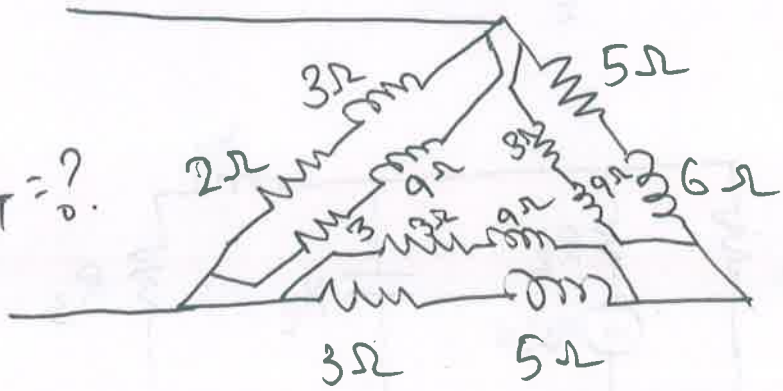


Converting the λ to Δ:

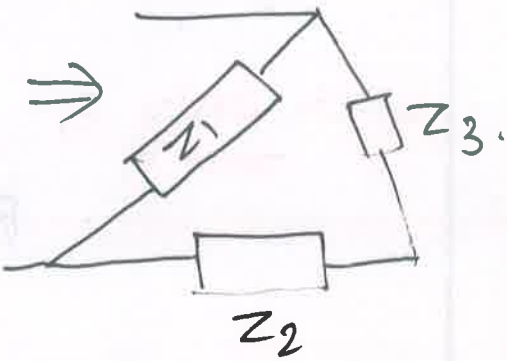
$$Z_\Delta = 3 Z_\lambda = 3[1 + j3] = \cancel{4+}\ 3 + 9j\ \Omega.$$

Circuit becomes:-

$Z_T = ?$



⇒



$$Z_3 = \frac{(3+9j)(5+6j)}{8+15j} = 2.19 + 3.76j\ \Omega.$$

$$Z_2 = \frac{(3+9j)(3+5j)}{6+14j} = 1.6 + 3.25j\ \Omega.$$

$$Z_1 = \frac{(3+9j)(2+3j)}{5+12j} = 1.29 + 2.29j$$

$$Z_T = Z_1 \parallel (Z_2 + Z_3)$$

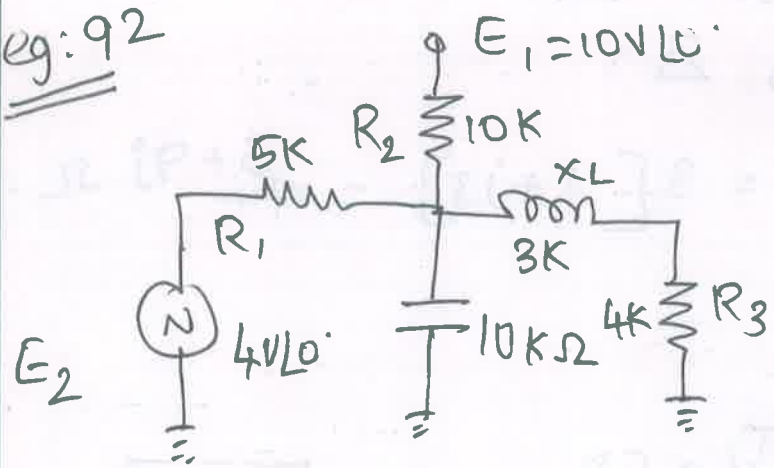
$$= (1.29 + 2.29j) \parallel \left[(1.6 + 3.25j) + (2.19 + 3.76j)\right]$$

$$Z_T = 0.96 + 1.7j\ \Omega.$$

Use calc:
- Complex mode
- store Z_1 in A
  Z_2 in B &
  Z_3 in C.

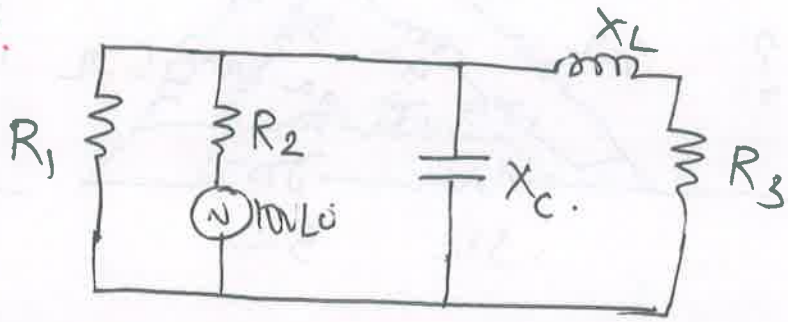
Superposition theorem:-

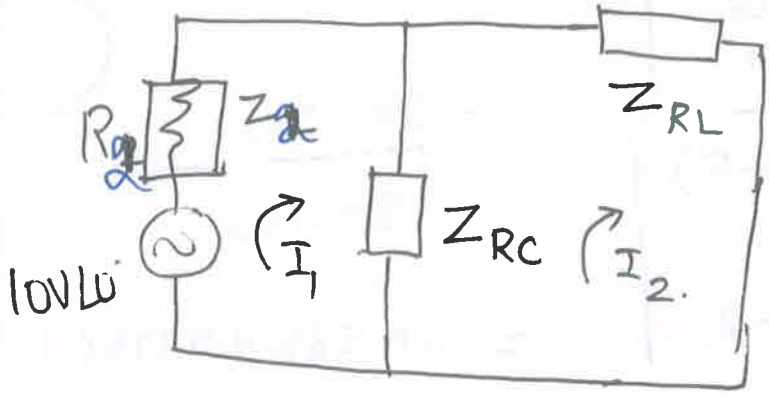
eg: 92



find V_{R_3}

With only E_1:





$$Z_2 = R_2 = 10K + j0 = 10K\angle 0^\circ$$

$$Z_{RC} = \frac{R_1 X_C}{R_1 + X_C} = \frac{(5\angle 0^\circ)(10\angle 90^\circ)}{5 - j10}$$

$$= \frac{50\angle 90^\circ}{11.18\angle -63.4^\circ} = 4.47\angle 153.4^\circ$$

$$Z_{RL} = R_3 + jX_L = 4K + j3K\,\Omega$$

$$= 4 \mp j2K\Omega$$

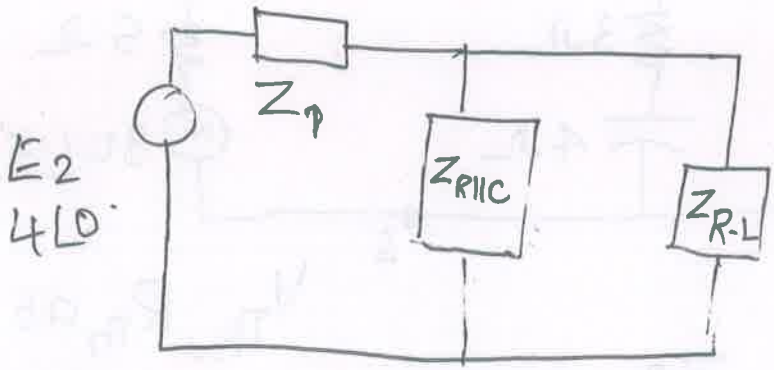
$$\cancel{I_2 = \frac{\begin{vmatrix} 14 \mp 2j & 10\angle 0 \\ -(4 \mp 2j) & 0 \end{vmatrix}}{\begin{vmatrix} (10+j0) + 4.0 \mp j2 & -(4 \mp j2) \\ -(4 \mp j2) & 8K \mp j \end{vmatrix}} = \frac{-40 \mp 20j}{\begin{vmatrix} 90 + 70i \end{vmatrix}}}$$

$$\cancel{I_2 = -0.384 + 0.076j = 0.39\angle 168\,A}$$

with only E_2:

$$I_2' = \frac{\begin{vmatrix} (14-2j)\times 10^3 & 10+j0 \\ (-4+2j)\times 10^3 & 0 \end{vmatrix}}{\begin{vmatrix} (14-2j)\times 10^3 & (-4+2j)\times 10^3 \\ (-4+2j)\times 10^3 & (8+j)\times 10^3 \end{vmatrix}}$$

$$= \frac{-40+20j}{102+14j} = (-0.36+0.24j)\,A$$



$$Z_1 = 5 + j0\,K\Omega$$

$$Z_{R_2 \| C} = \frac{(10)(-j10)}{10 - j10} = 5 - 5j\,K\Omega$$

$$Z_{R\text{-}L} = 4 + 3j\,K\Omega$$

$$I_2'' = \frac{\begin{vmatrix} 10-j5\,k\Omega & 4\angle 0^\circ \\ -(5-5j)\,k\Omega & 0 \end{vmatrix}}{\begin{vmatrix} 10-j5\,k\Omega & -(5-5j)\,k\Omega \\ -(5-5j)\,k\Omega & 9-2j\,k\Omega \end{vmatrix}} = \frac{-20+20j}{80-15j}$$

$$= -0.286 + 0.196j\ A$$

$$I_2 = I_2' + I_2'' = (-0.36 + 0.24j) + (-0.286 + 0.196j)$$

$$I_2 = -0.646 + 0.436j$$

$$I_2 = 0.779\angle 145.98^\circ\ A.$$

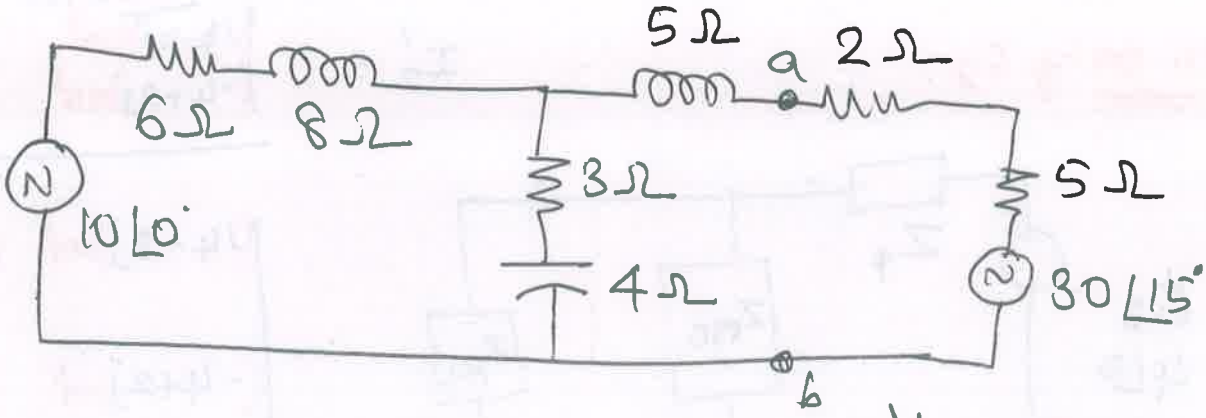
~~Thev~~

$$V_{R_3} = I_2 \times R_3 = 0.779\angle 145.98 \times 4k$$

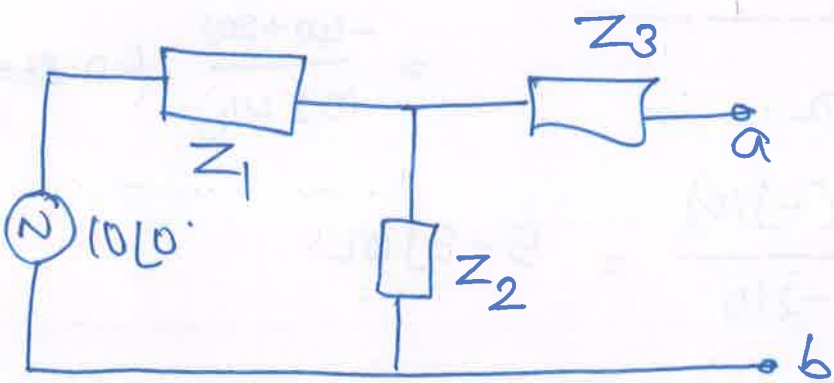
$$V_{R_3} = 3116\angle 145.98\ V.$$

Thevenin's Theorem:

eg:93



V_{Th}, R_{Th} at ab



$Z_1 = 6 + j8\,\Omega.$

$Z_2 = 3 - 4j\,\Omega$

$Z_3 = 0 + j5\,\Omega$

$$Z_{Th} = Z_3 + \frac{Z_1 Z_2}{Z_1 + Z_2}$$

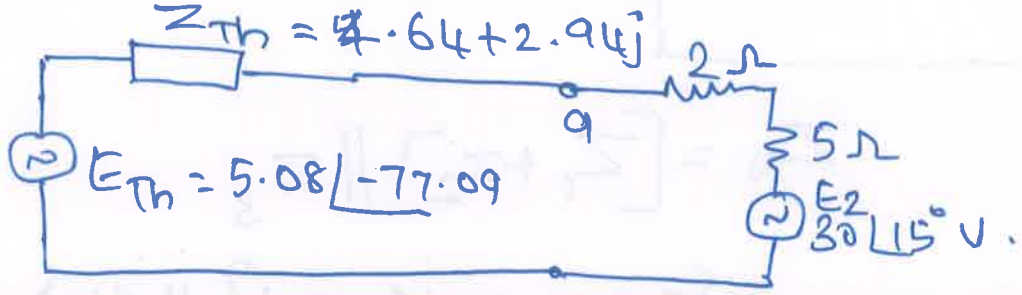
$$Z_{Th} = (0+j5) + \frac{(6+j8)(3-4j)}{9+4j} = 4.64 + 2.94j\,\Omega$$

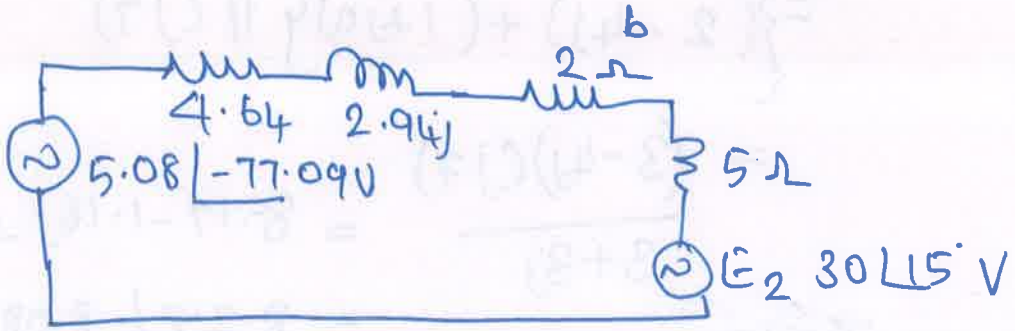
$$Z_{Th} = 5.5\angle 32.36^\circ\,\Omega.$$

$$E_{Th} = E\frac{Z_2}{Z_2+Z_1} = (10+j0)\frac{(3-4j)}{(3-4j)+(6+8j)}$$

$$E_{Th} = 1.13 - 4.94j$$

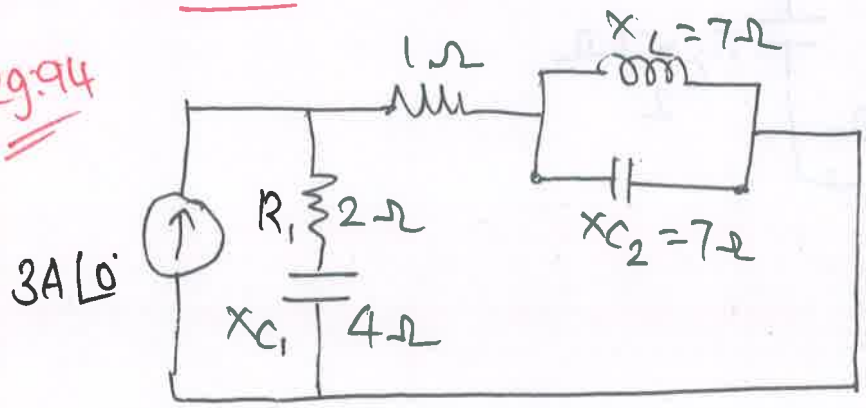
$$E_{Th} = 5.08\,V\angle -77.09^\circ\,V.$$



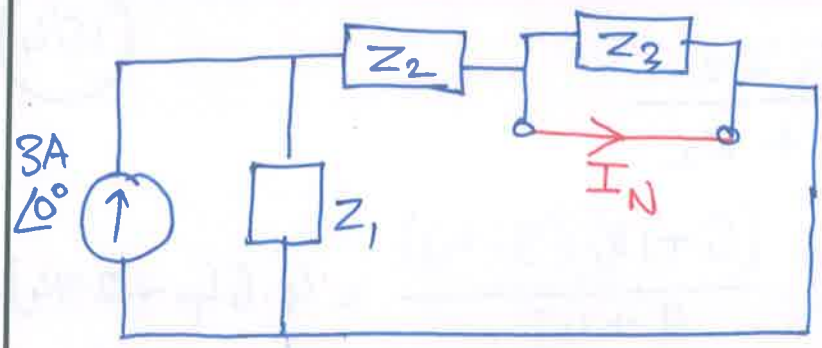


Norton's Theorem:-

29.94



Norton equivalent ~~to~~ at X_{C_2}.



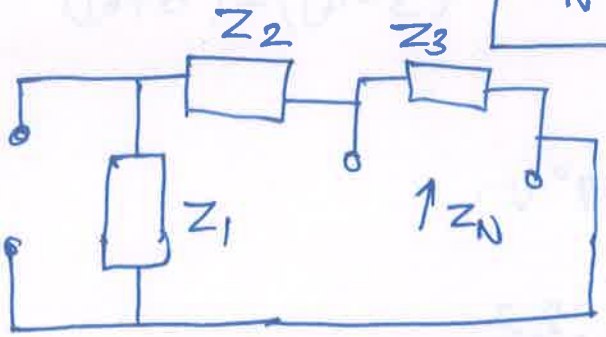
$Z_1 = 2-4j$

$Z_2 = 1+j0$; $Z_3 = j7\,\Omega$

$$I_N = (3+j0)\frac{Z_1}{Z_1+Z_2} = (3+j0)\frac{2-4j}{3-4j}$$

$$\boxed{I_N = 2.64-0.48j\ A.}$$

$$I_N = 2.68\angle{-10.3}\ A.$$

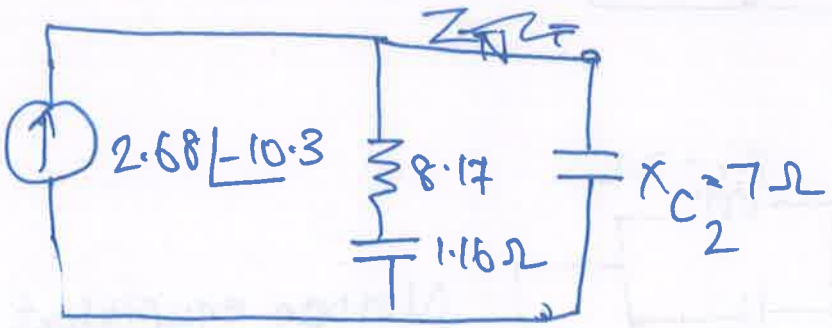


$$Z_N = [Z_1+Z_2] \parallel Z_3$$

$$= \{(2-4j)+(1+j0)\} \parallel (j7)$$

$$= \frac{(3-4j)(j7)}{3+3j} = 8.17-1.16j\,\Omega$$

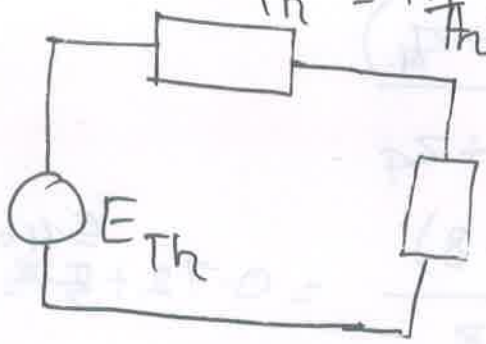
$$= 8.25\angle{-8.08}\,\Omega.$$



Maximum Power transfer theorem:-

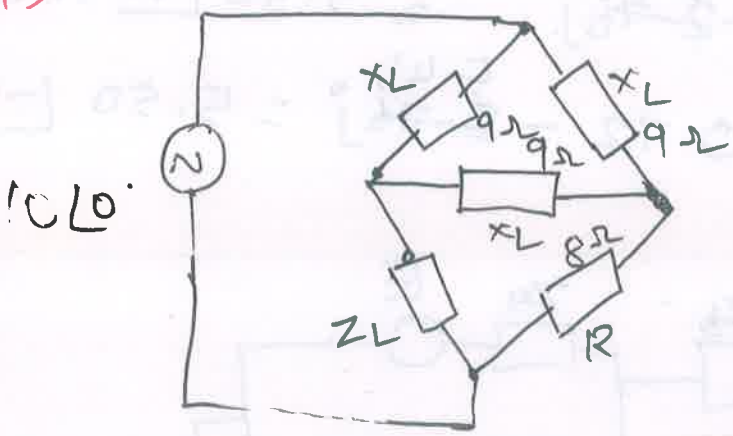
Maximum power will be delivered to the load when the load impedance is equal to the complex conjugate of the Thevenin impedance across the terminals.

$$Z_{Th} = R_{Th} \pm jX_{Th} = Z_{Th} \angle \theta_{Th}$$



$$Z_L = R_L \mp jX_L = Z_L \angle -\theta_{Th}$$

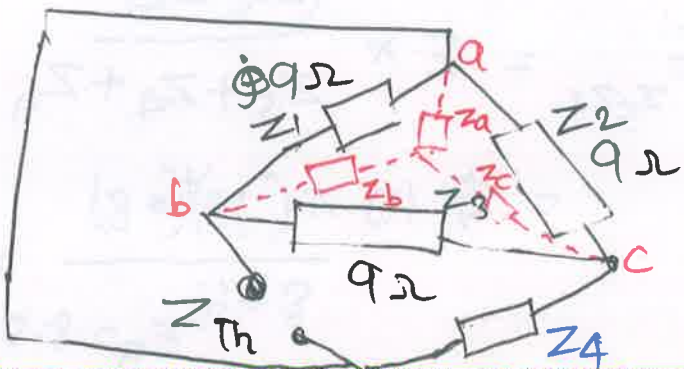
eg:-95



Find Z_L for ~~MP~~ Maxm. Power.

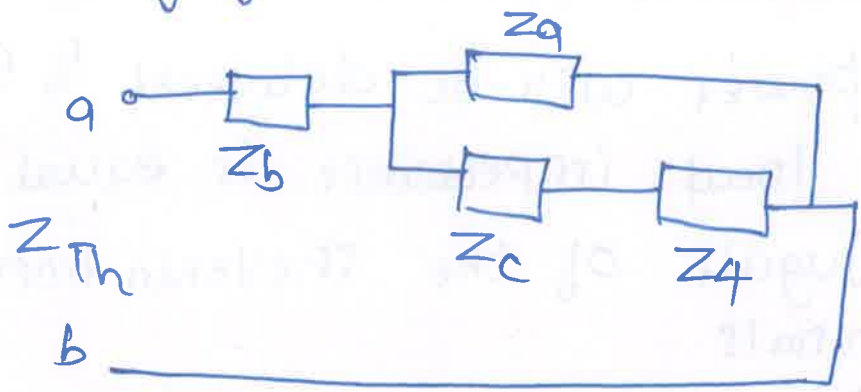
$$Z_1 = jX_L = j9\,\Omega = Z_2 = Z_3$$
$$Z_4 = R = 8\,\Omega$$

Z_{Th}:



$$Z_\Delta = 3Z_Y$$
$$Z_a = Z_b = Z_c = \frac{j9}{3}$$
$$Z_Y = 3j\,\Omega.$$

Rearranging the circuit,



Z_{Th}

$$Z_{Th} = Z_b + \frac{Z_a(Z_c+Z_4)}{Z_a+Z_c+Z_4}$$

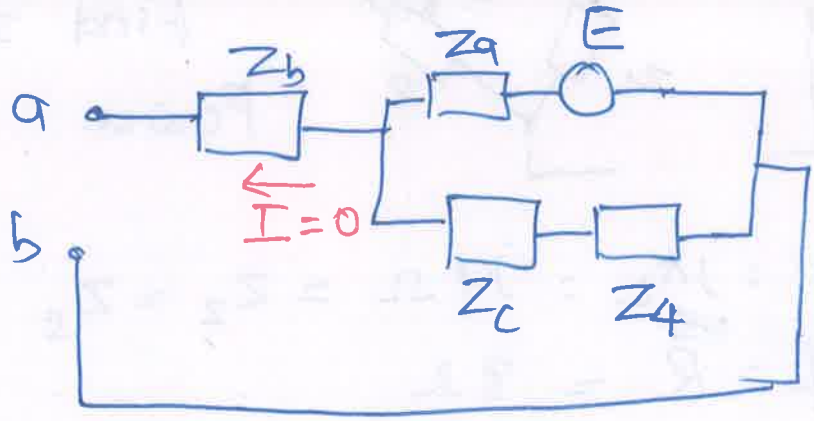
$$= j3 + \frac{j3(j3+8)}{j3+j3+8} = 0.72 + \cancel{5.76}\,5.46j$$

$$Z_{Th} = 0.72 + \cancel{2.76}\,5.46j = 5.50\angle 82.48^\circ\ \Omega$$

$$Z_L = Z_{Th}^* = 0.72 - \cancel{2.76}\,5.46j = 5.50\angle -82.48\ \Omega$$

E_{Th}:

$E_{Th} =$



By using VDR,

$$E_{Th} = E_{Z_cZ_4} = E \times \frac{(Z_c+Z_4)}{Z_c+Z_4+Z_a}$$

$$= 10 \times \frac{(j3)(8)}{8+6j}$$

$8.2 - 2.4j\ \Omega$

$= \cancel{14.4+19.2j}$

$E_{Th} = 8.54\angle -16.3\ V$

$$P_{max} = \frac{E_{Th}^2}{4R_{Th}} = \frac{8.54^2}{4(0.72)} = 25.32W$$

Power in ac circuits:

Power in dc is product of V & I.

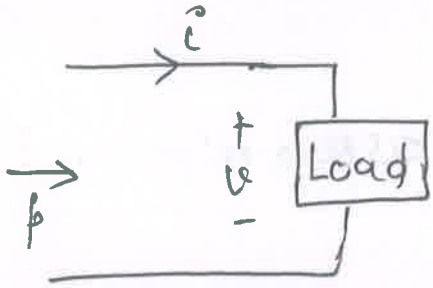
$P_{dc} = VI$ Watts.

In ac,

$P_{ac} = v\,i$

where $v = V_m \sin(\omega t + \theta)$

$i = I_m \sin(\omega t)$



$\theta = 0$, if load is R

$\theta = 90$, if " is L

$\theta = -90$ " " " C.

$$P_{ac} = V_m I_m \sin(\omega t + \theta) \cdot \sin(\omega t).$$

$$\cancel{\sin(A+B) = \sin A \cos B + \cos A \sin B}$$

$$2\sin A \sin B = \cos(A-B) - \cos(A+B)$$

$$\Rightarrow \cancel{P_{ac} = V_m I_m \{\sin^2 \omega t \cos\theta + \cos\omega t \sin\omega t \sin\theta\}}$$

$$P_{ac} = \frac{V_m I_m}{2}\left[\cos\theta - \cos(2\omega t + \theta)\right]$$

Average P_{ac} : $\text{avg}\left\{\frac{V_m I_m}{2}\left[\cos\theta - \cos(2\omega t + \theta)\right]\right\}$

$$= \text{avg}\left(\frac{V_m I_m}{2}\cos\theta\right) - \underbrace{\text{avg}\left(\frac{V_m I_m}{2}\cos(2\omega t + \theta)\right)}_{\to 0}$$

$$\boxed{P_{av} = \frac{V_m I_m}{2}\cos\theta}$$

$$P_{av} = \frac{V_m I_m}{2} \cos\theta$$

$$= \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos\theta = V_{rms} I_{rms} \cos\theta$$

For resistive circuit:

$$\theta = 0 \Rightarrow \cos\theta = 1$$

$$P_{av} = V_{rms} I_{rms}$$

For Inductive circuit:

$$\theta = 90^\circ \Rightarrow \cos\theta = 0$$

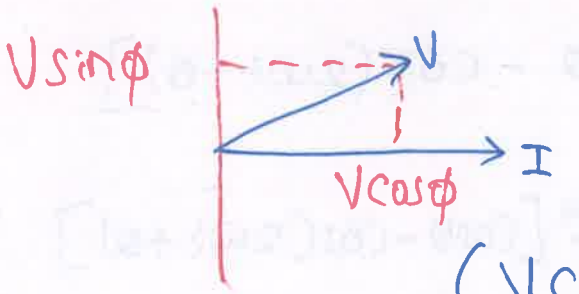
$$P_{av} = 0$$

For capacitive circuit.

$$\theta = -90^\circ \Rightarrow \cos\theta = 0$$

$$P_{av} = 0.$$

Power factor, $\cos\theta$:

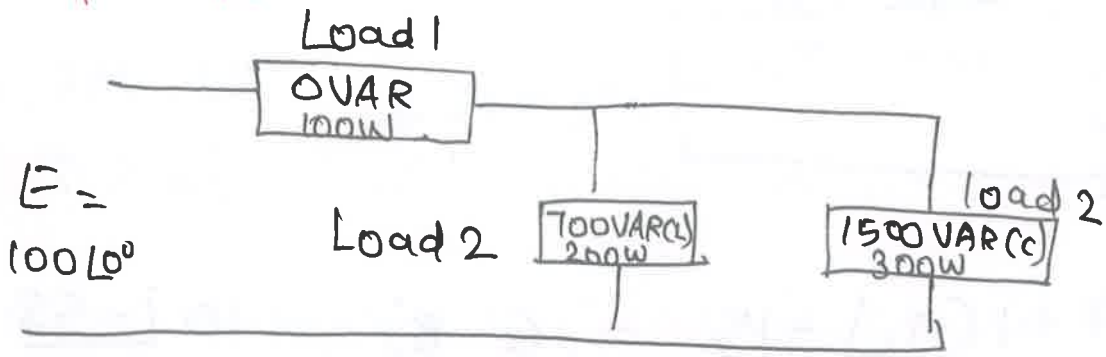


$(V\cos\phi)I$ - resistive power

$(V\sin\phi)I$ - reactive power.

Ex: 96, 97 → midsem problems.

Example: 98



find total W, VAR, VA, F_p; I in phasor.

load	W	VAR	VA = S = $\sqrt{P^2+Q^2}$
1	100	0	S = 100
2	200	700 (L)	$\sqrt{200^2+700^2}$ = 728 VA
3	300	1500 (C)	$\sqrt{300^2+1500^2}$ = 1529.71 VA
	P = 600 W	Q = 800 (C)	$\sqrt{600^2+800^2}$ = 1000

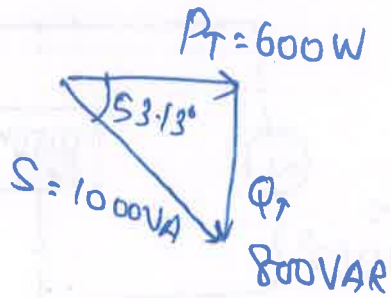
$1000 \neq 100+728+1529.$

$$F_p = \frac{P}{S} = \frac{600W}{1000VA} = 0.6 \text{ leading (C)}$$

I = ?

$$S = VI = 1000VA$$

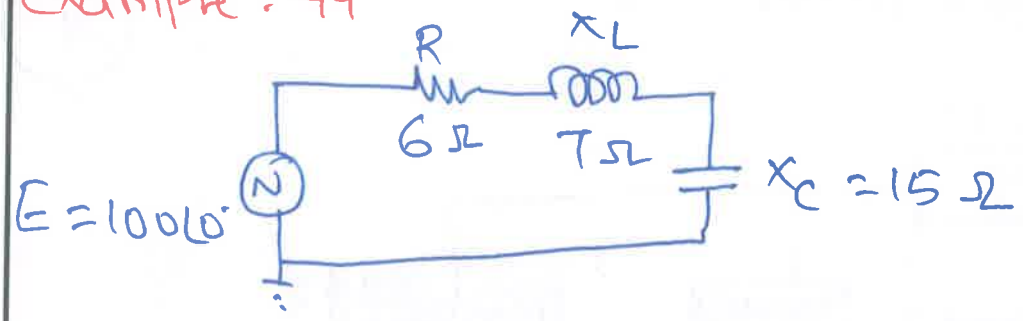
$$I = \frac{1000}{100} = 10A.$$



I in phasor.

$$I = 10\angle 53.13° A.$$

example: 99



W, VAR, VA, $F_p = ?$

$$Z_T = R + j(X_L) - jX_C = 6 - 8j = 10\angle -53.13^\circ\ \Omega$$

$$I = \frac{E}{Z_T} = \frac{100}{6-8j} = \frac{100}{10\angle -53.13^\circ} = 10\angle 53.13^\circ\ A$$

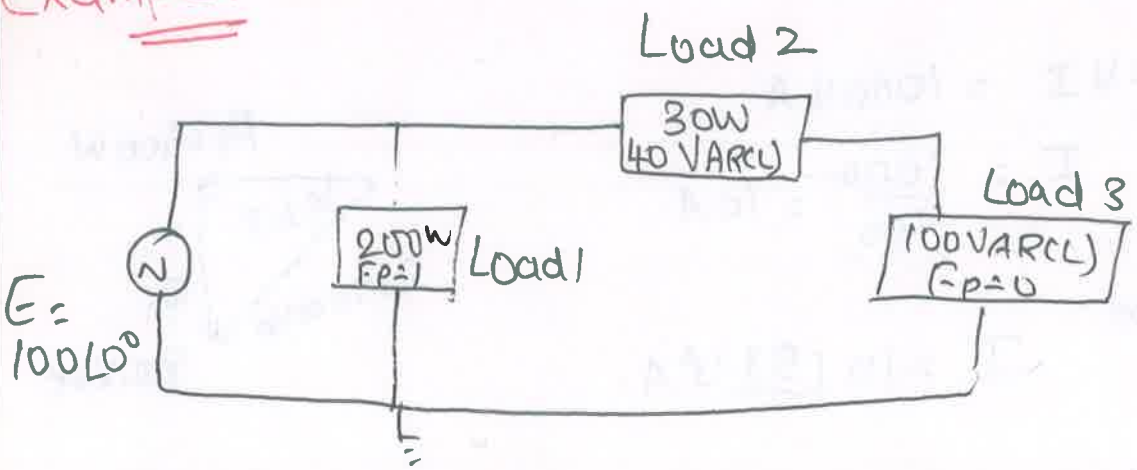
$$P = EI\cos\theta = 100 \times 10 \cos 53.13 = 600\,W.$$

$$Q = EI\sin\theta = 100 \times 10 \sin 53.13 = 800\,VAR.$$

$$S = EI = 1000\,VA$$

$$F_p = \frac{P}{S} = \frac{600}{1000} = 0.6 \text{ leading (c)}.$$

example: 100:



$$P = 230\,W.$$

$$Q = 0 + 40(L) + 100(L) = 140\,VAR(L)$$

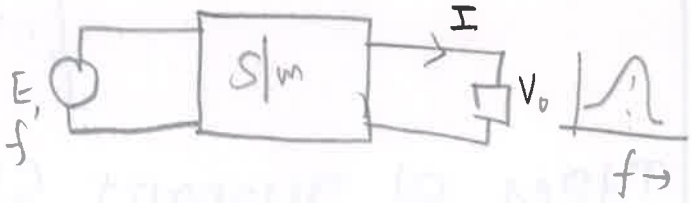
$$S = \sqrt{230^2 + 140^2} = 269.26\,VA.$$

$$F_p = \frac{P}{S} = \frac{230}{269.26} = 0.854 \text{ lag (L)}$$
$$= 31.35^\circ.$$

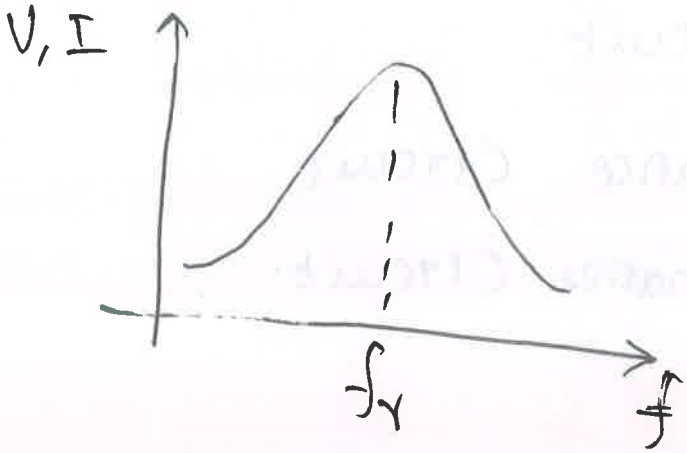
$$I_s = \frac{S}{E} = \frac{269.26\,VA}{100V} = 2.6926\,A.$$

$$I = 2.69 \angle -31.35^\circ$$

Resonance



eg: radio or television receiver.



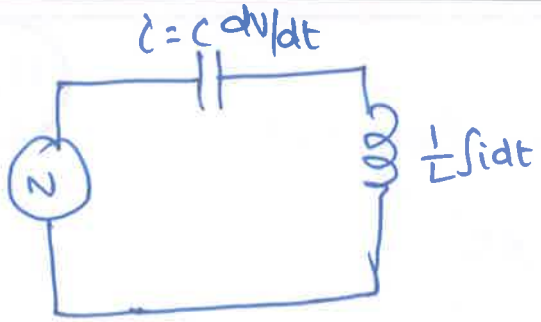
State of resonance:

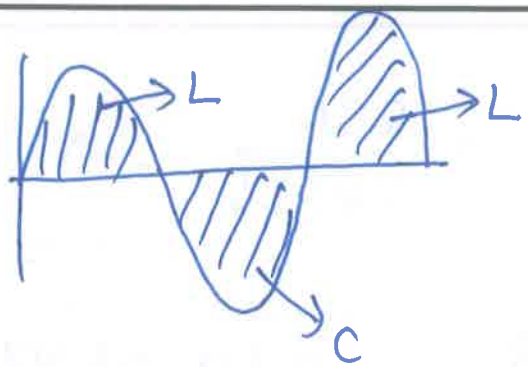
When the response of a s/m occurs maximum at a frequency, a state of resonance occurs.

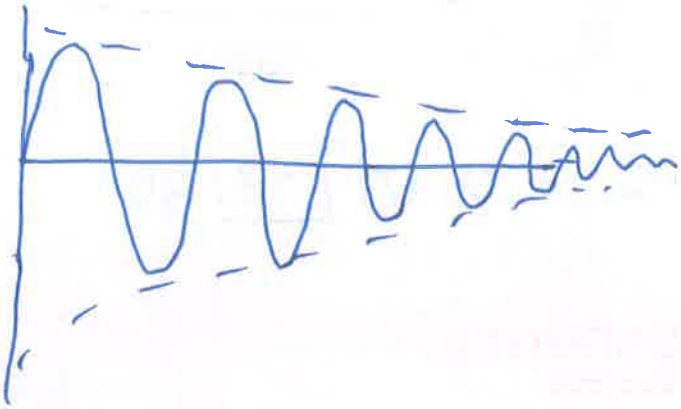
Resonant electric s/m contains at R L & C.

During resonance, energy absorbed by one reactive element is equal to energy released by another reactive element.

Natural frequency: Resonance is a process in which an obj's nat. freq. is amplified by an identical freq.





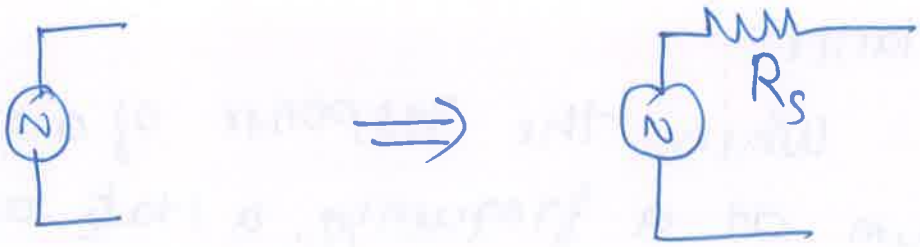


Types of Resonant Circuit:

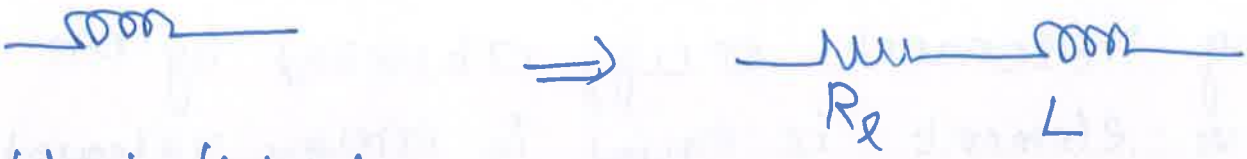
→ Series resonance circuit

→ parallel resonance circuit.

Series Resonant circuit:

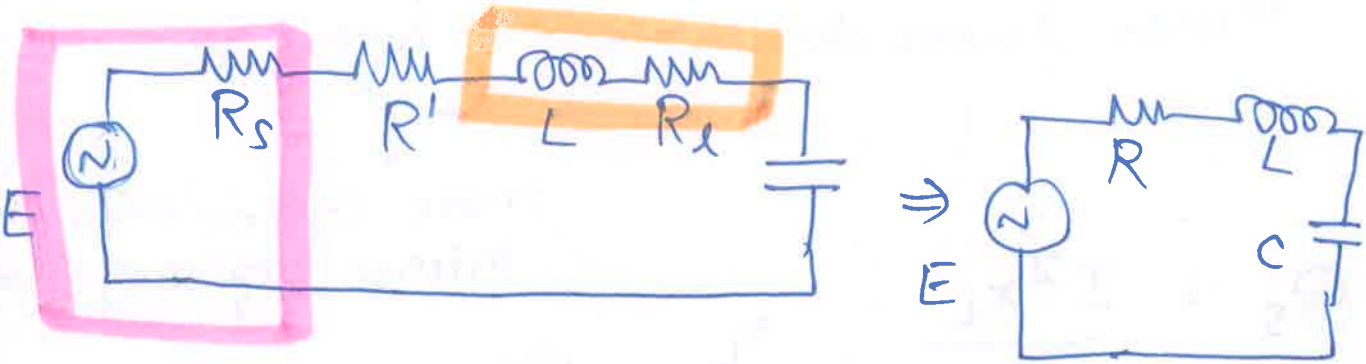


Ideal Source practical source.



Ideal Inductor practical Inductor

Actual circuit for series resonance:



$$R = R_s + R' + R_l$$

$$Z = R + jX_L - jX_C$$

$$= R + j(X_L - X_C)$$

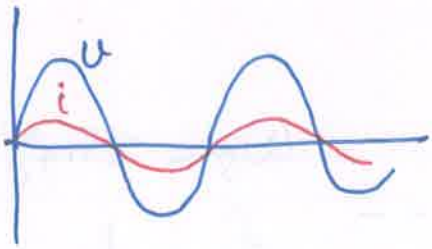
At resonance,

$$X_L - X_C = 0.$$

$$X_L = X_C \qquad 2\pi f L = \frac{1}{2\pi f C}$$

$$f = \frac{1}{2\pi\sqrt{LC}}$$

$\Rightarrow$ $Z_r = R$ [purely a resistive circuit)



$\Rightarrow \theta = 0^\circ$

$\Rightarrow \cos\theta = 1$

$F_p = 1$

Quality factor: Q_s

$$Q_s = \frac{\text{reactive power}}{\text{average power}}$$

Indicates how much energy is placed in storage

Compared to dissipated.

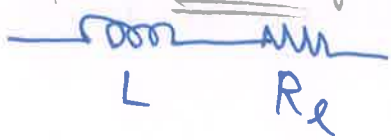
lower power dissipation → higher Q

↓

more concentrated and intense region of resonance

$$Q_s = \frac{I^2 x_L}{I^2 R} = \frac{x_L}{R} = \frac{\omega_s L}{R}$$

Q factor of a coil:-



$$Q_s = Q_{coil} = Q_l = \frac{x_L}{R_l}$$

Q_l

Skin effect → affects R_l

Capacitance b/w coil turns increase

$f \rightarrow$ log scale.

Considering,

$$Q_s = \frac{\omega_s L}{R} \qquad \omega_s = 2\pi f_c$$

$$f_c = \frac{1}{2\pi\sqrt{LC}}$$

$$Q_s = \frac{1}{\sqrt{LC}} \cdot \frac{L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

Q_s in terms of circuit parameters.

$Q_s > 1$